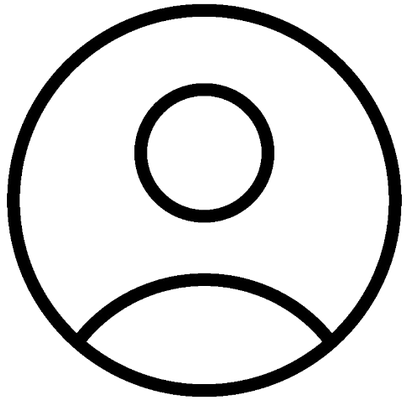
Multi AI Agent Systems with crewAI

Joao Moura





DeepLearning.AI



Role Playing

Focus

Tools

Cooperation

Guardrails

Memory

Multi AI Agent Systems with crewAI

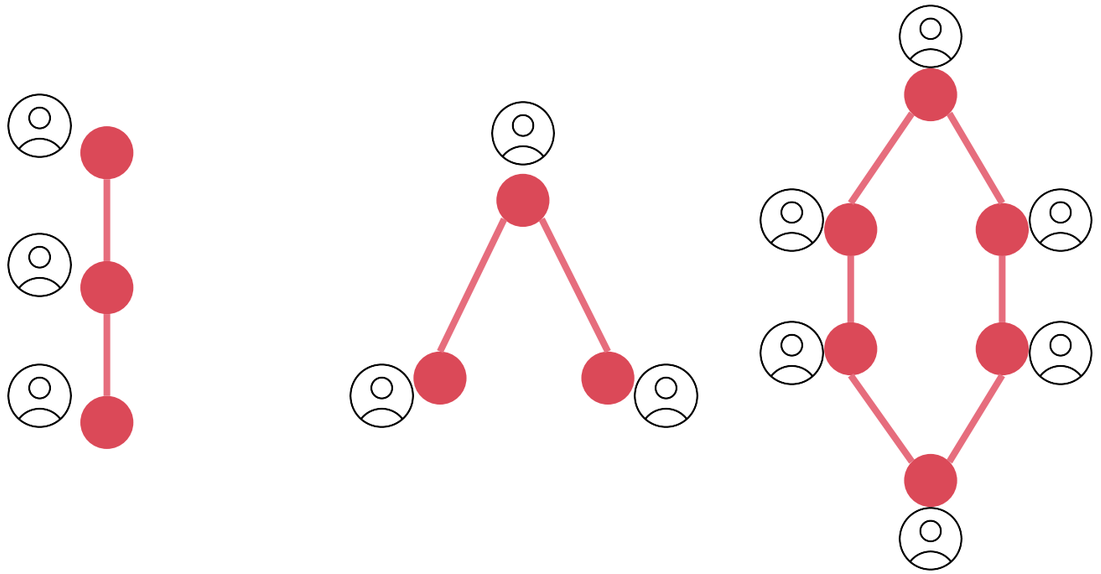
Overview





DeepLearning.AI

Multi-Agent Collaboration



What we will cover in this course

- Overview
  - Research & write a technical article
- Multi Agents core concepts
  - Customer support AI Agents
- Using Tools at Scale
  - Customer outreach campaign
- Tasks and Processes
  - Event planning
- Agentic Collaboration
  - Financial analysis
- Build a complete Crew
  - Tailor-made resumes
- Next Steps

What will we build in this course?

Full Stack Engineer

APPLY FOR THIS JOB

Colombia

AI Fund – AI Fund / Full Time /

Job Description:

We are looking for Full Stack Engineers to produce scalable software solutions for our portfolio companies. You'll be part of a cross-functional and will be responsible for the full software development life cycle, from conception to deployment.

As a Full Stack Developer, you should be comfortable around both front-end and back-end coding languages, development frameworks and third-party libraries. You should also be a team player with a knack for visual design and utility.

What You Will be Doing:

- Work with development teams and product managers to ideate software solutions
- Build the front-end of applications through appealing visual design
- Develop and manage well-functioning databases and applications
- Write effective APIs
- Test software to ensure responsiveness and efficiency
- Troubleshoot, debug and upgrade software
- Create security and data protection settings
- Build features and applications

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What You Must Bring:

Must Haves:

- Proven experience as a Full Stack Developer or similar role
- Experience developing desktop and mobile applications
- Familiarity with common stacks
- Knowledge of multiple front-end languages and libraries (e.g. HTML/ CSS, JavaScript, XML, jQuery)
- Knowledge of multiple object-oriented languages (e.g. Java, Python) and multiple frameworks (e.g. Angular, React, Node.js)
- Familiarity with databases (e.g. MySQL, MongoDB), web servers (e.g. Apache) and UI/UX design
- Excellent communication skills and the ability to have in-depth technical discussions with both the engineering team and business people
- Excellent English language skills
- Self-starter and comfort working in an early-stage environment
- Strong project management and organizational skills

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Profile

Noah Williams is a distinguished Software Engineering Leader with an 18-year tenure in the technology industry, where he has excelled in leading both remote and in-office engineering teams. His expertise spans across software development, process innovation, and enhancing team collaboration. He is highly proficient in programming languages such as Ruby, Python, JavaScript, TypeScript, and Elixir, alongside deep expertise in various frameworks. Noah's significant experience in data science and machine learning has enabled him to spearhead successful deployments of scalable AI solutions and innovative data model development.

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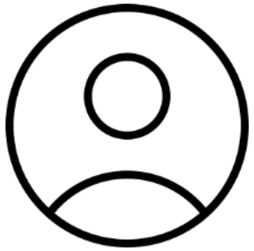
BetCraft: CTO — 2013 - 2015

- Led the technology strategy post-Series A funding, directly reporting to the board and guiding the company through a phase of significant technological advancement and network expansion.
- His strategic initiatives and partnerships significantly improved platform performance and expanded the company's market reach.
- Helped build his initial product using both React and Angular and got pretty good at it.

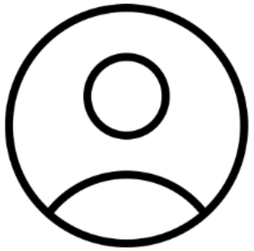
InnovPet: Founder & CEO (remote) — 2019 - 2022

- Noah founded InnovPet, a startup focused on innovative IoT solutions for pet care, including a revolutionary GPS tracking collar that significantly enhanced pet safety and owner peace of mind.
- He was responsible for overseeing product development from concept through execution, working closely with engineering teams and marketing partners to ensure a successful market entry.
- Successfully set up an advisory board, established production facilities overseas, and navigated the company through a successful initial funding phase, showcasing his leadership and entrepreneurial acumen.
- Built the initial version of the product leveraging MongoDB

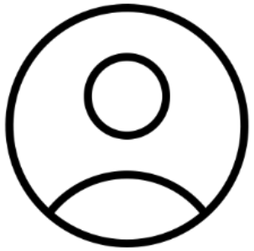
Agents



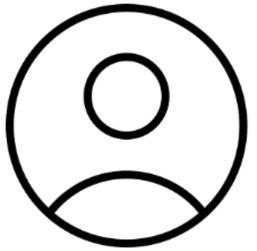
Tech Job Researcher



Personal Profiler for Engineers

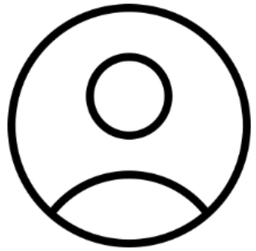


Resume Strategist for Engineers

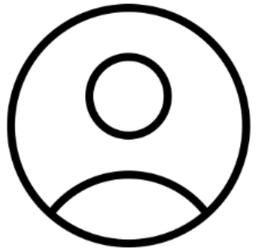


Engineering Interview Preparer

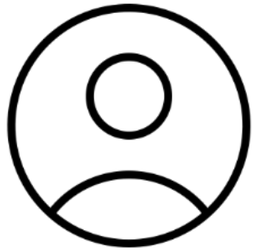
Agents



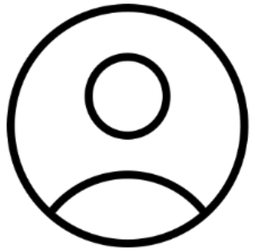
Tech Job Researcher



Personal Profiler for Engineers



Resume Strategist for Engineers



Engineering Interview Preparer

- Search the internet
- Read websites
- Read resume
- Perform RAG (Retrieval Augmented Generation) on resume

Profile

Noah Williams is an accomplished Full Stack Developer with 18 years of experience in software engineering. He is proficient in multiple programming languages including JavaScript, Python, and Ruby, and has extensive experience with front-end frameworks such as React and Angular, as well as back-end development using Node.js. His expertise also extends to UI/UX design, mastering HTML, CSS, and JavaScript to create responsive and visually appealing interfaces. Additionally, Noah has a strong background in managing databases like MySQL and MongoDB, ensuring data integrity and security across all applications.

(original)

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re-written by an AI agent

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BetCraft: CTO — 2013 - 2015

- Managed full-stack development during a critical growth phase, enhancing platform capabilities using Angular and React for the front-end, and improving backend services.

InnovPet: Founder & CEO (remote) — 2019 - 2022

- Developed an innovative IoT solution for pet care, from database management with MongoDB to front-end user interface design, ensuring a seamless user experience.

EliteDevs: Engineering Manager (remote) — 2018 - 2019

- Implemented new product features using Ruby on Rails and React, focusing on both server-side and client-side code.

Key Interview Questions and Talking Points for Noah Williams

Technical Skills and Experience:

1. **Front-end Development:**
   - **Question:** "Could you walk us through your experience using front-end frameworks like React and Angular in your projects at BetCraft and EliteDevs?"
   - **Talking Point:** Discuss the challenges and outcomes of integrating these frameworks into existing systems.
2. **Back-end and API Development:**
   - **Question:** "At DriveAI, you developed a central API. What were some of the key considerations and challenges you faced in optimizing backend logic and database efficiency?"
   - **Talking Point:** Highlight the impact of these optimizations on system performance and user satisfaction.
3. **Database Management:**
   - **Question:** "You have significant experience with MySQL and MongoDB. Can you share how you ensured data integrity and security in your role at InnovPet and DataKernel?"
   - **Talking Point:** Focus on the strategies employed to manage large datasets and ensure high availability and security.

Project Management and Leadership:

4. **Strategic Initiatives:**

   - **Question:** "As Director of Software Engineering at DataKernel, you've led strategic initiatives integrating AI technologies. How did you align these technologies with business goals?"
   - **Talking Point:** Emphasize the role of AI in enhancing product capabilities and the challenges of implementing such advanced technologies.

5. **Team Leadership and Culture Building:**
   - **Question:** "Managing remote teams across multiple time zones presents unique challenges. How did you foster a culture of transparency and high performance at DataKernel?"
   - **Talking Point:** Discuss methods for motivating teams and maintaining productivity in a remote setting.

What can I do with Multi-Agents?

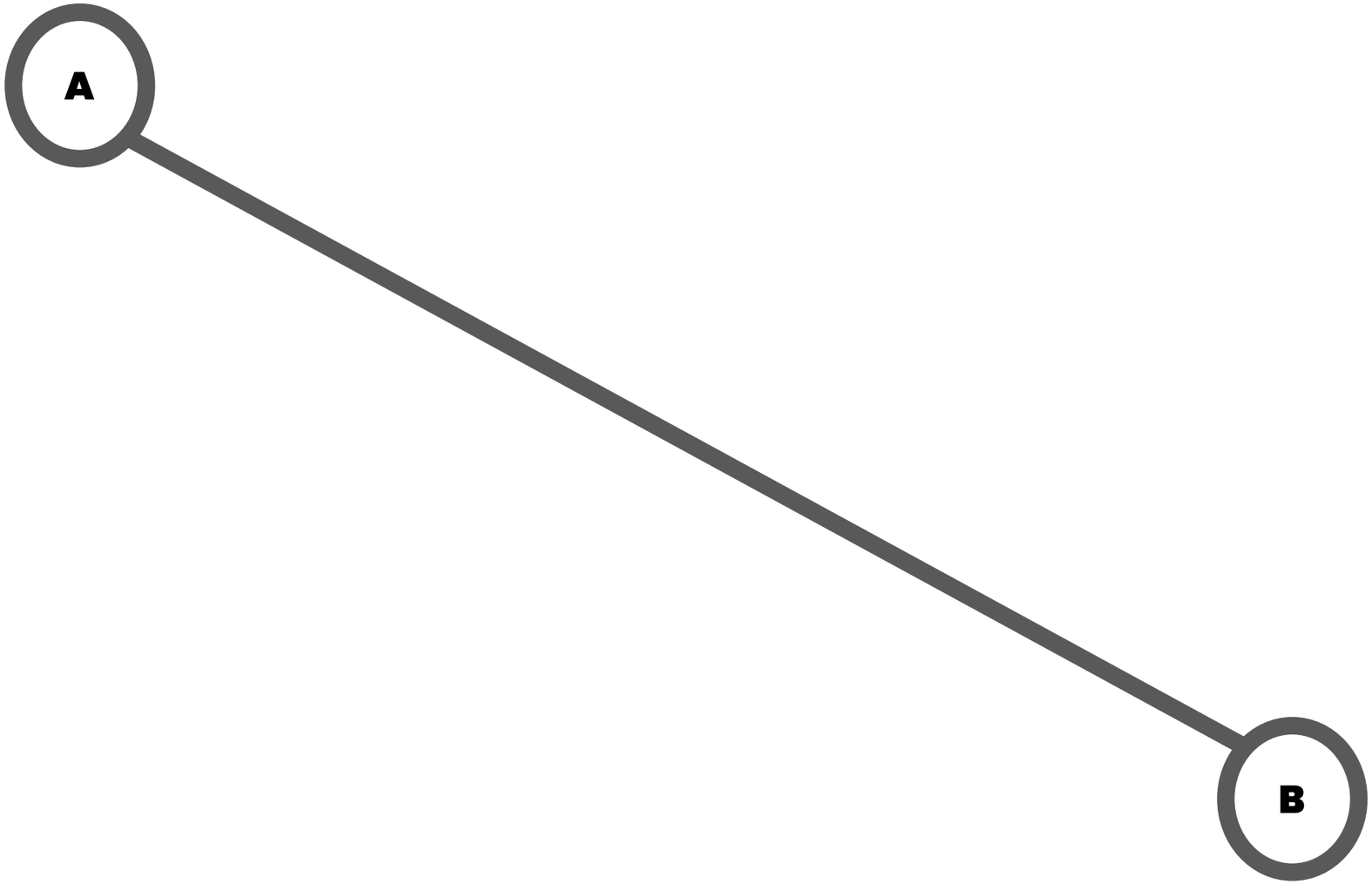
What can I do with Multi-Agents?

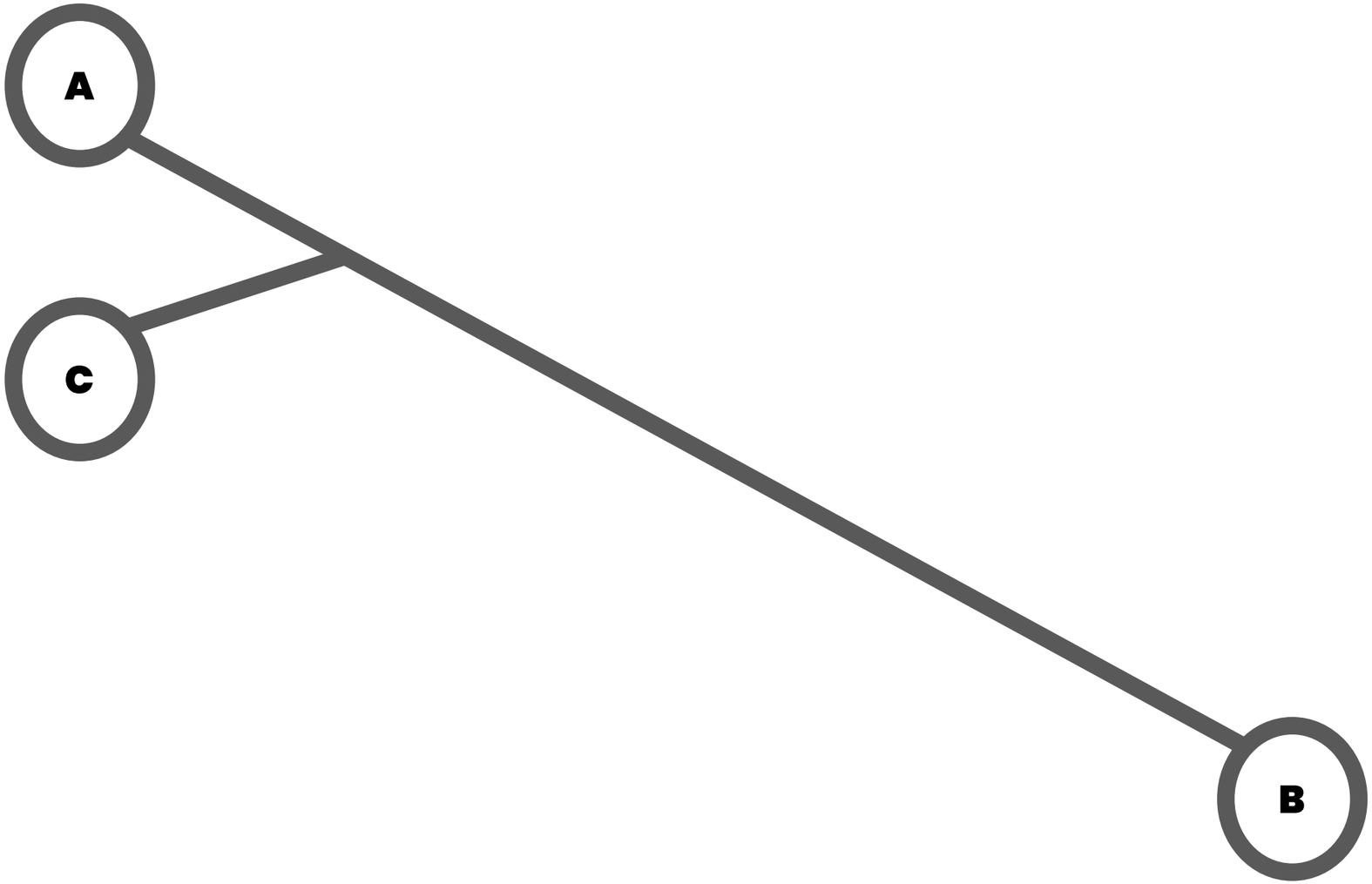
- Tailor job applications for each job description
- Generate reports on many fronts (ex. financials, HR, development, etc..)
- Help research and qualify leads
- Documentation generation
- Marketing analysis, research and creation
- Monitoring manufacturing data

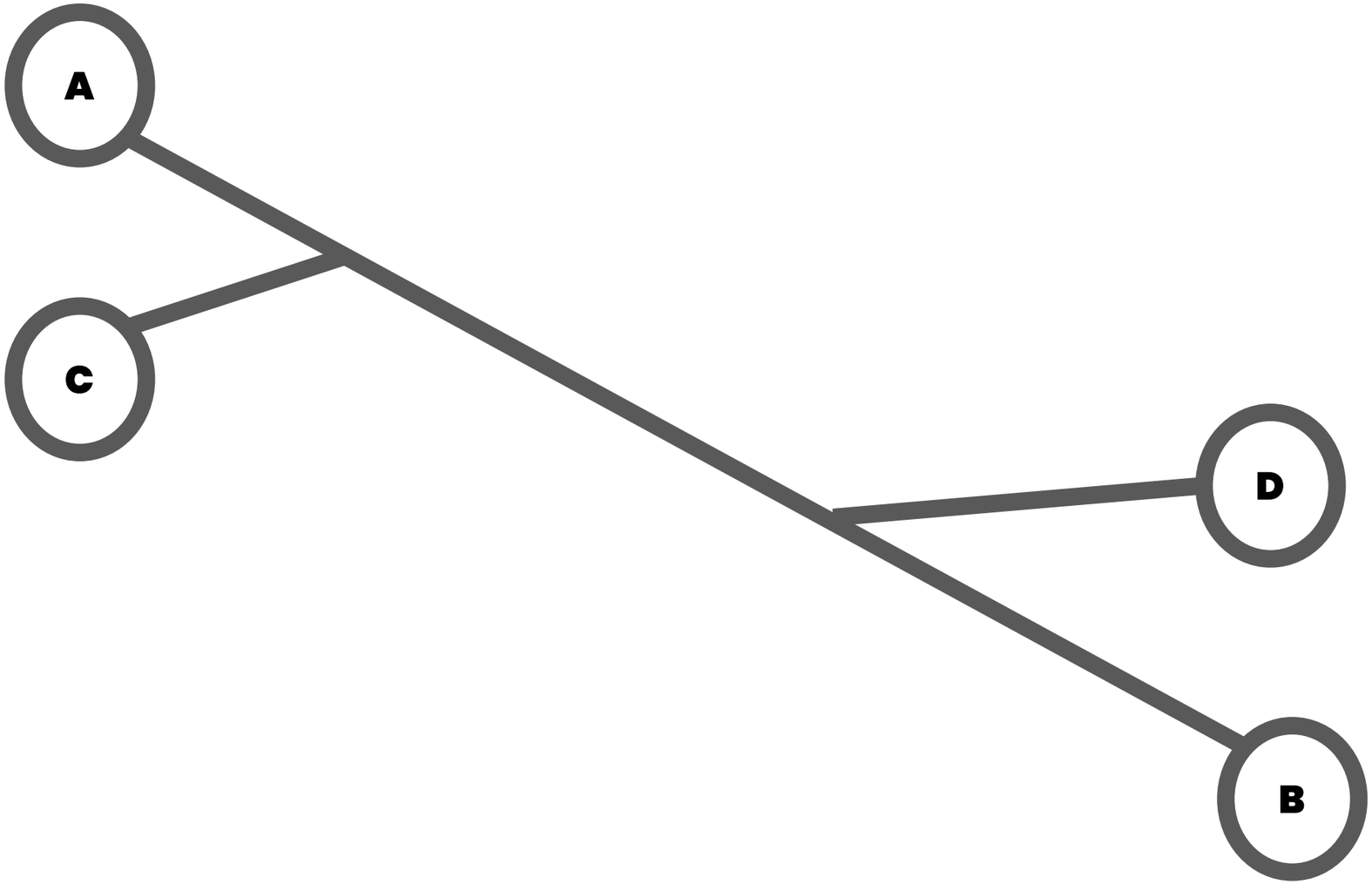
What we will cover in this course

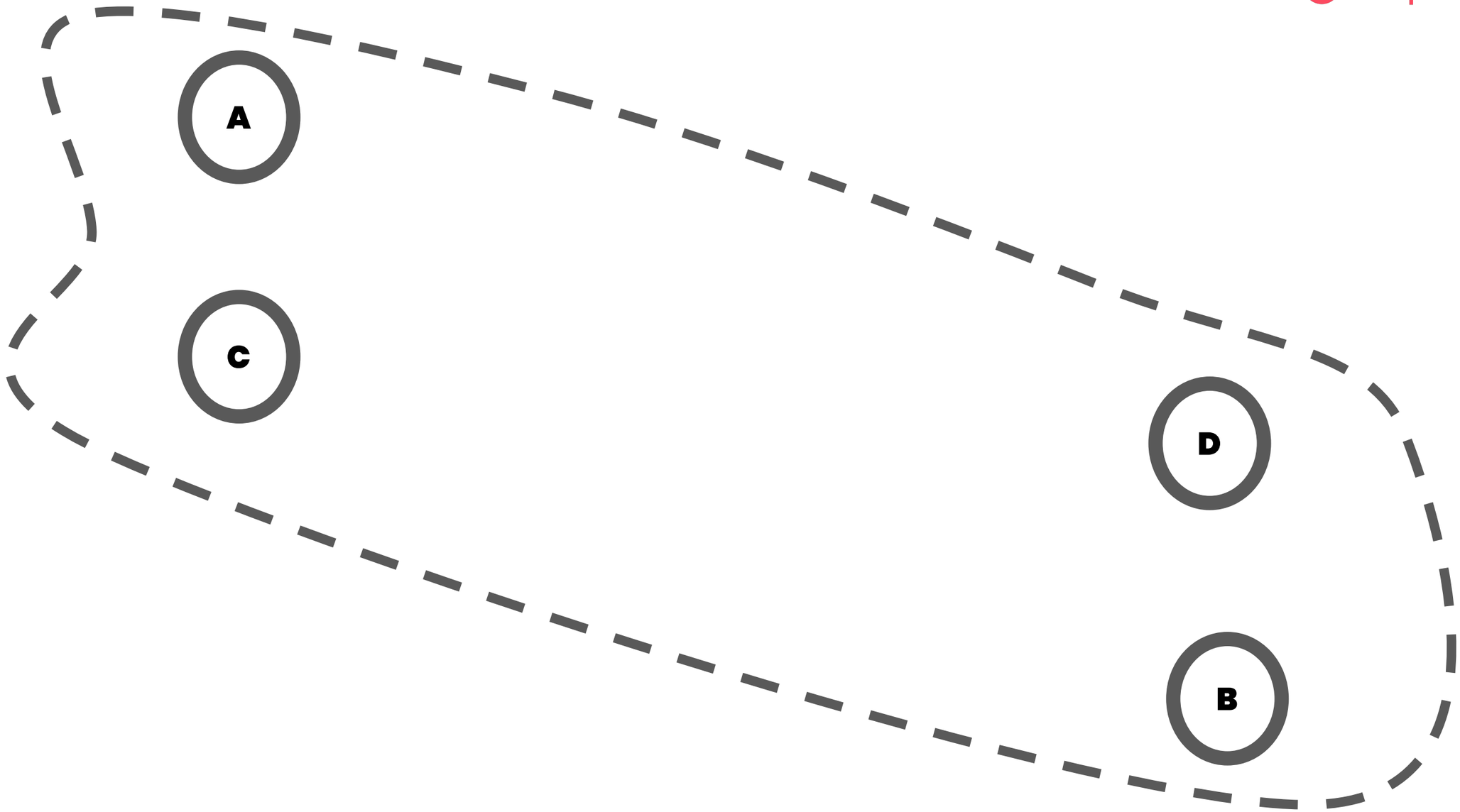
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What is Agentic Automation?


A
B


A
C
B

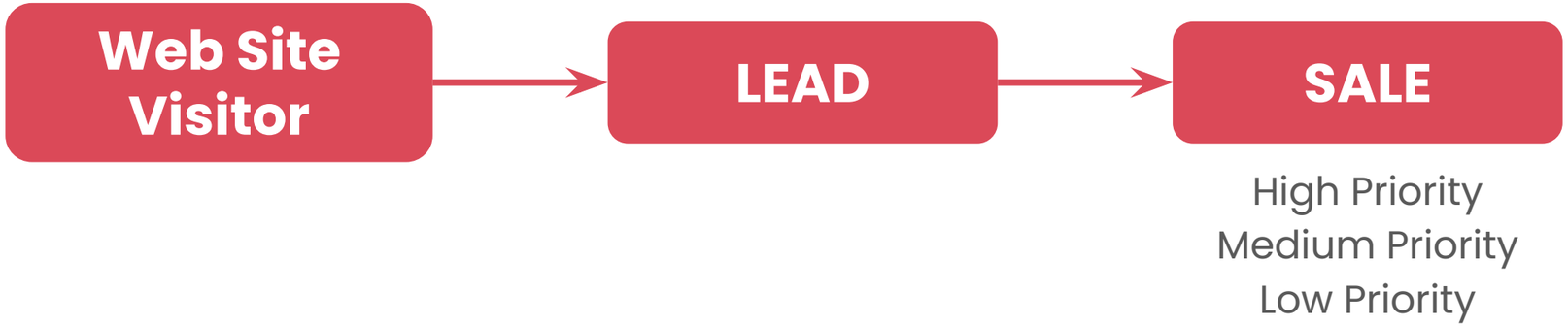

A
C
D
B


A
C
D
B

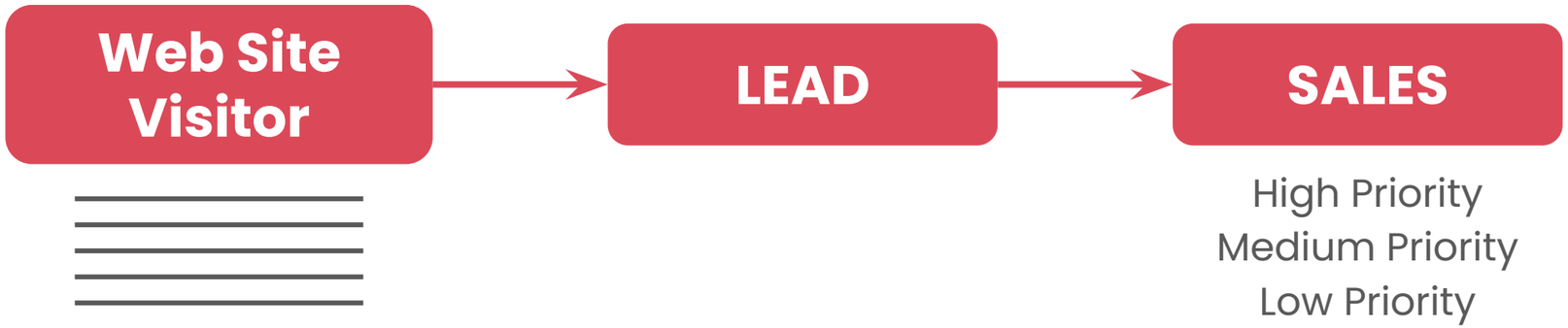
Fuzzy inputs, transformations, outputs

	Traditional Software Development	**AI Software Development**
Inputs	")	
Transformations		
Outputs		
Notes		

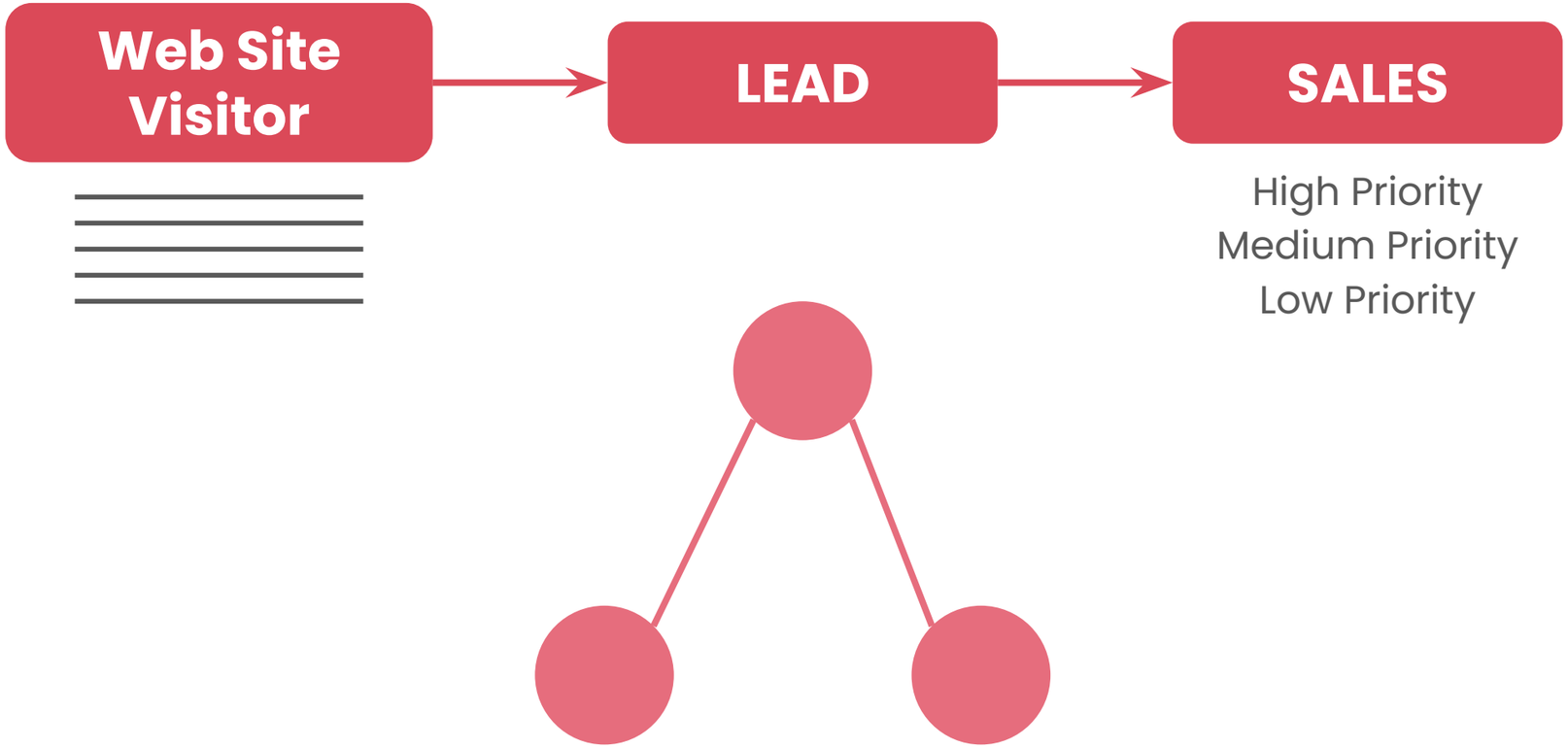
Data Collection and Analysis



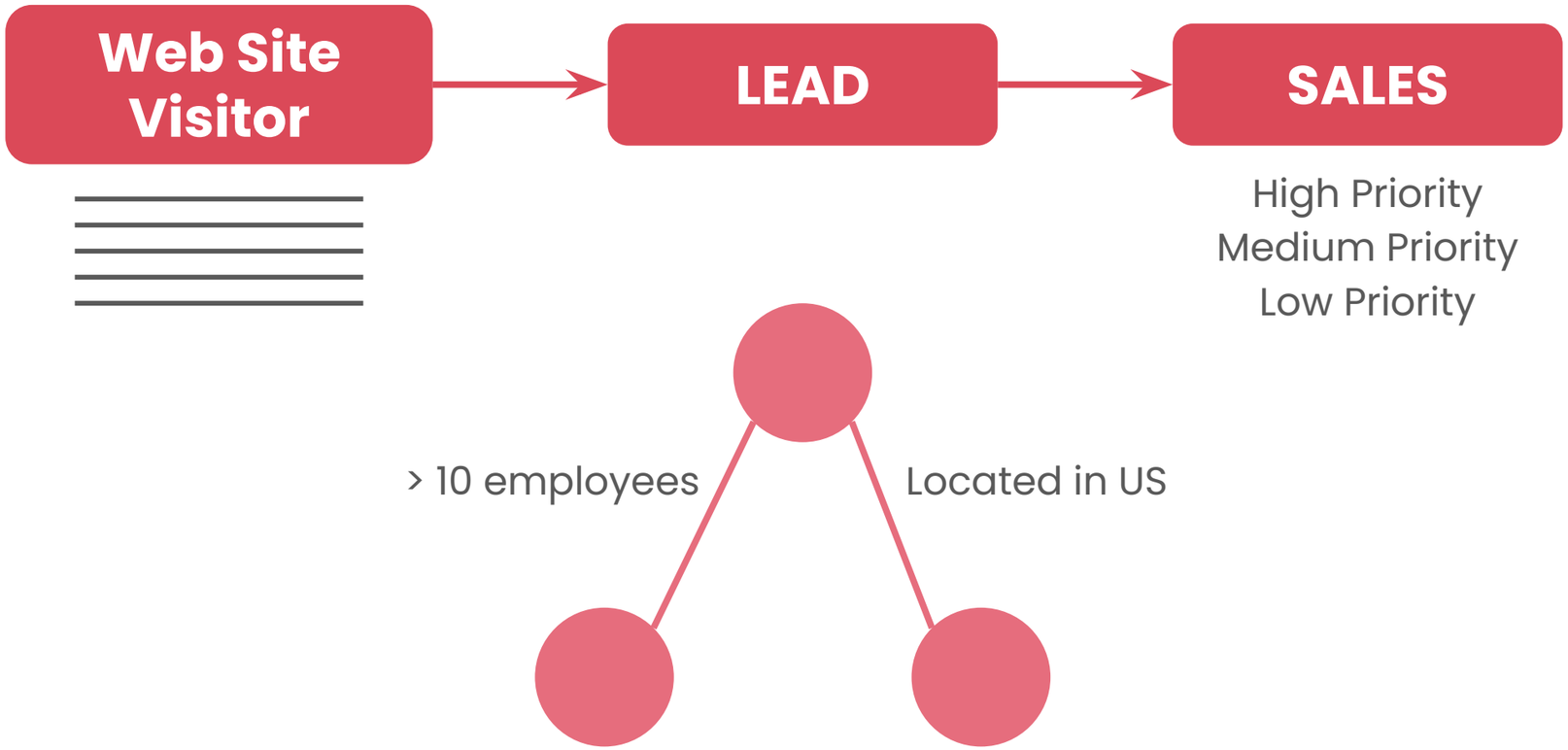
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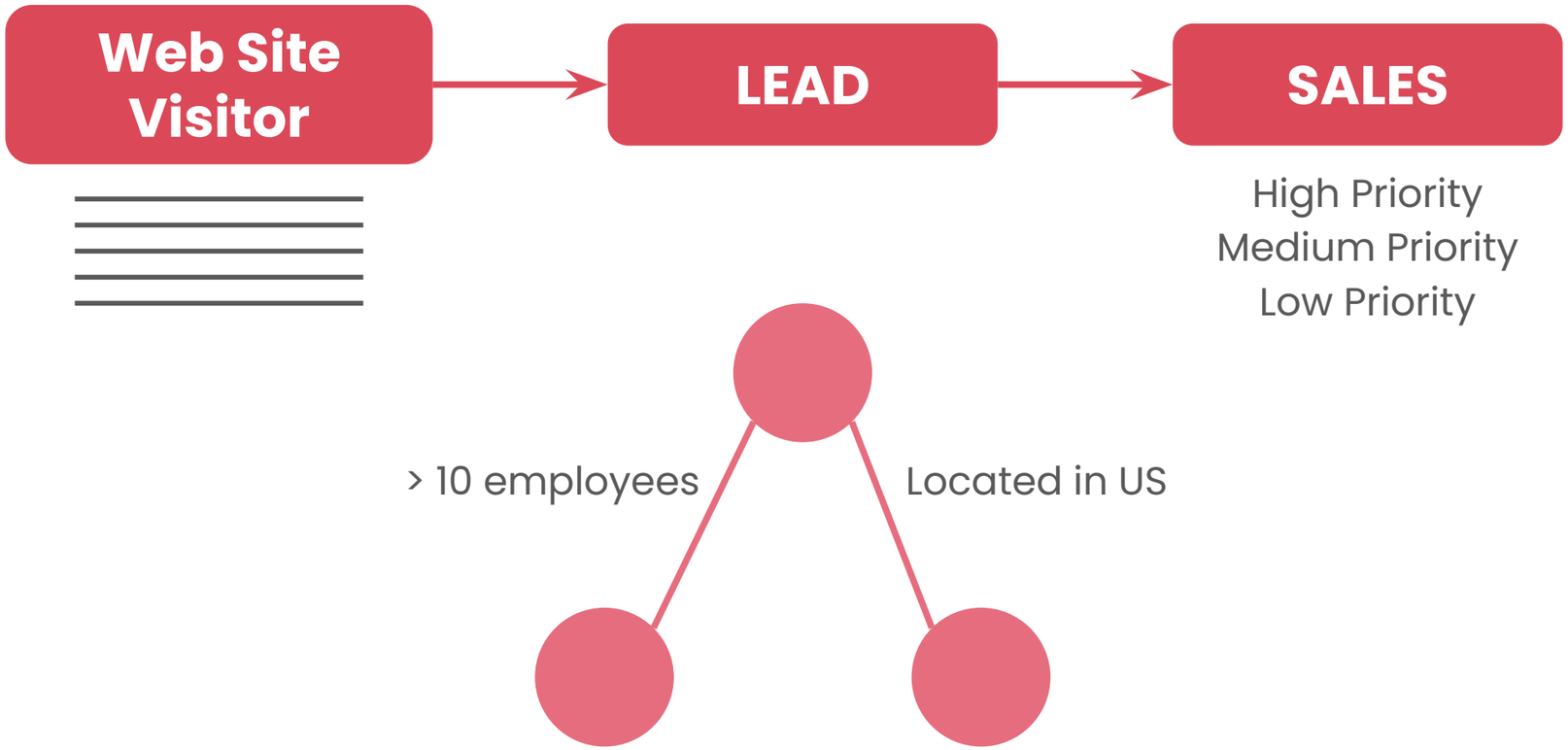
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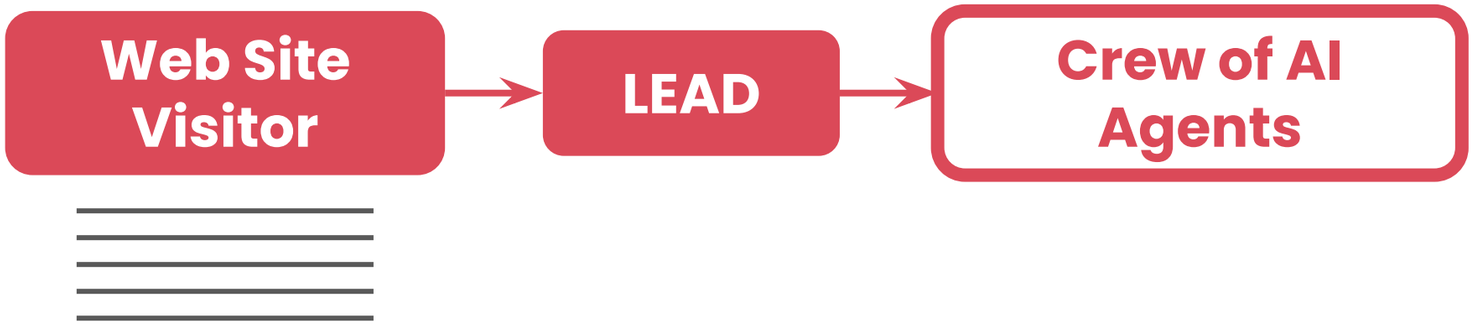
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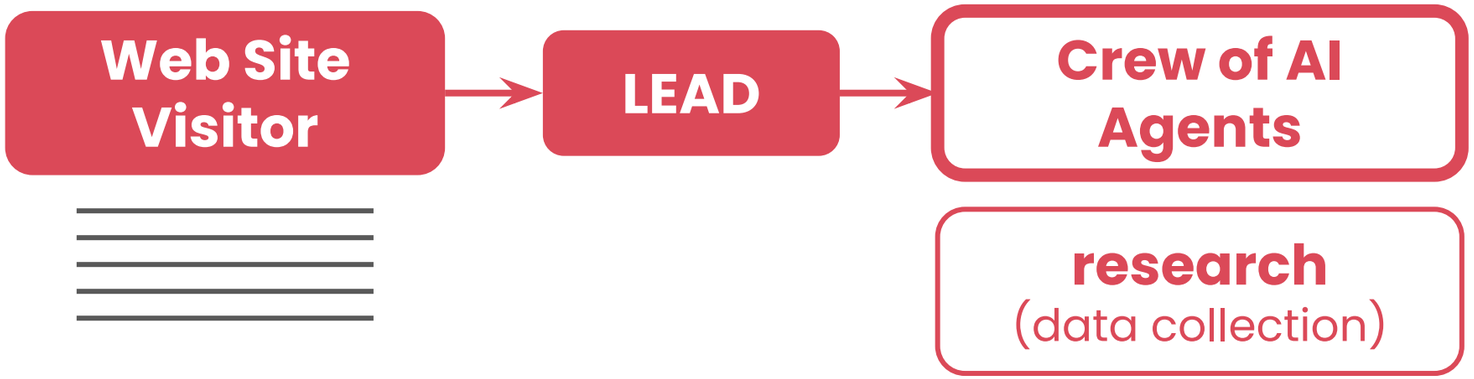
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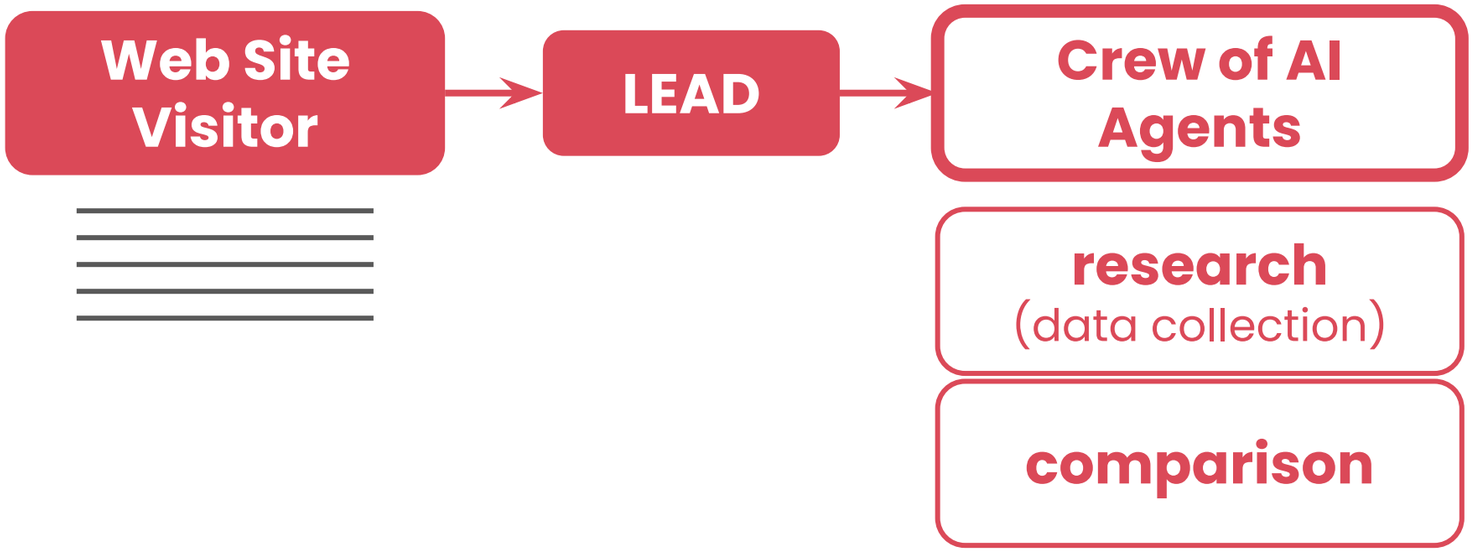
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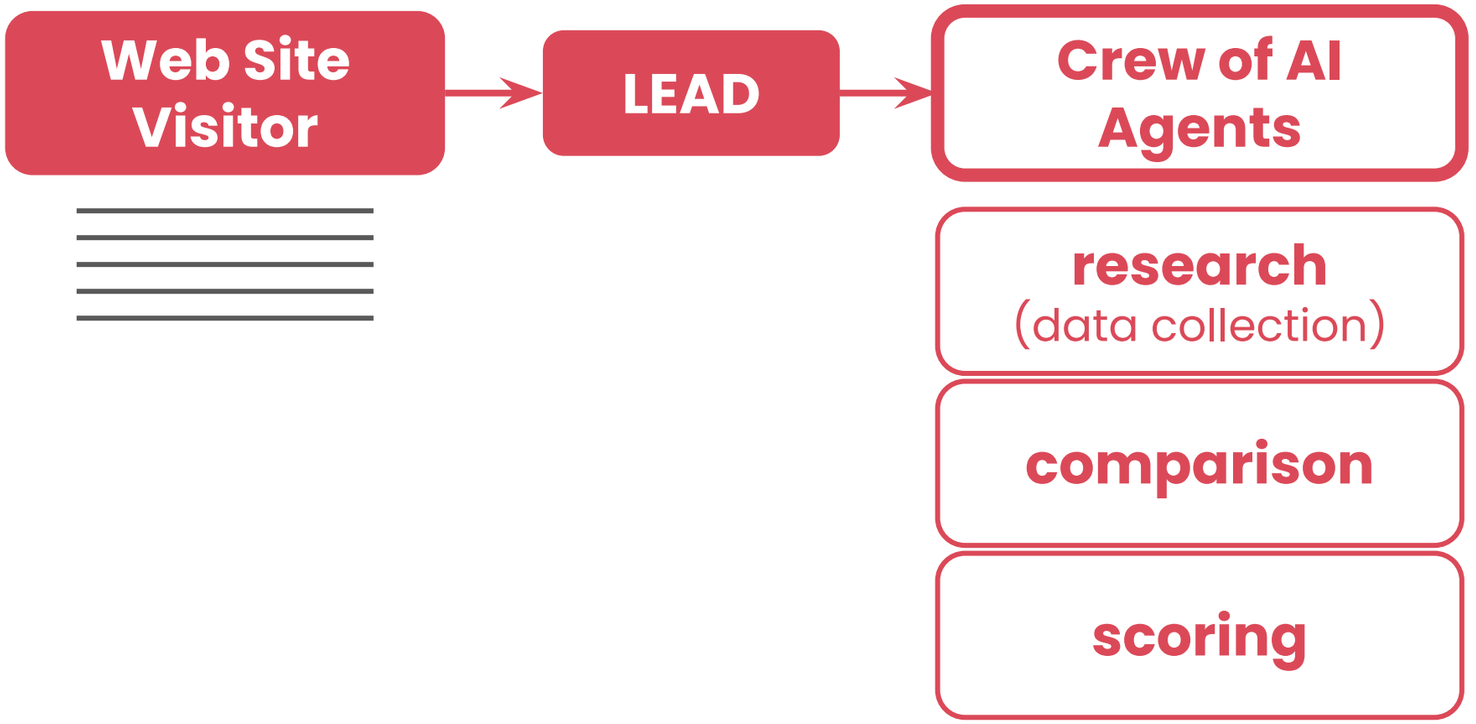
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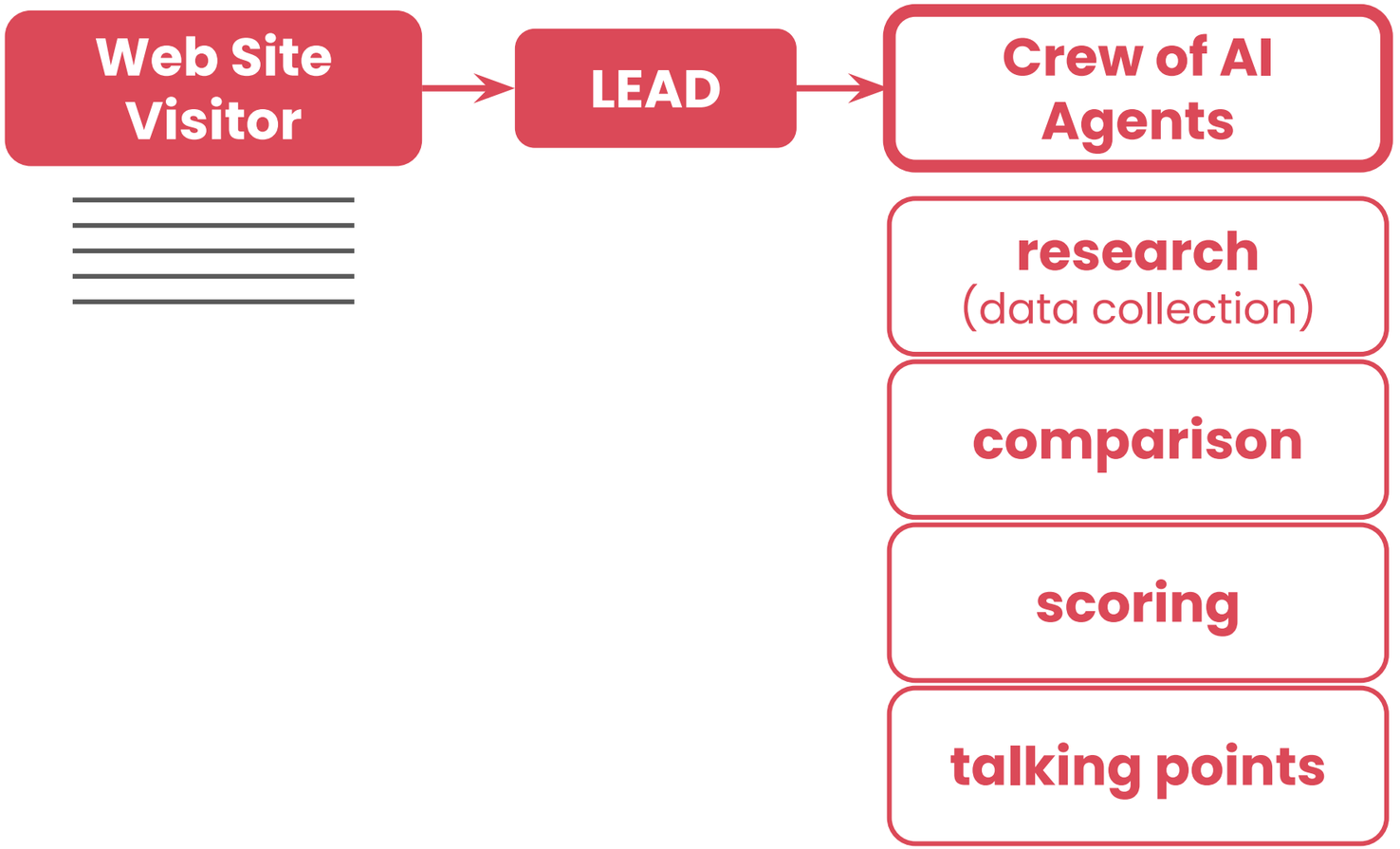
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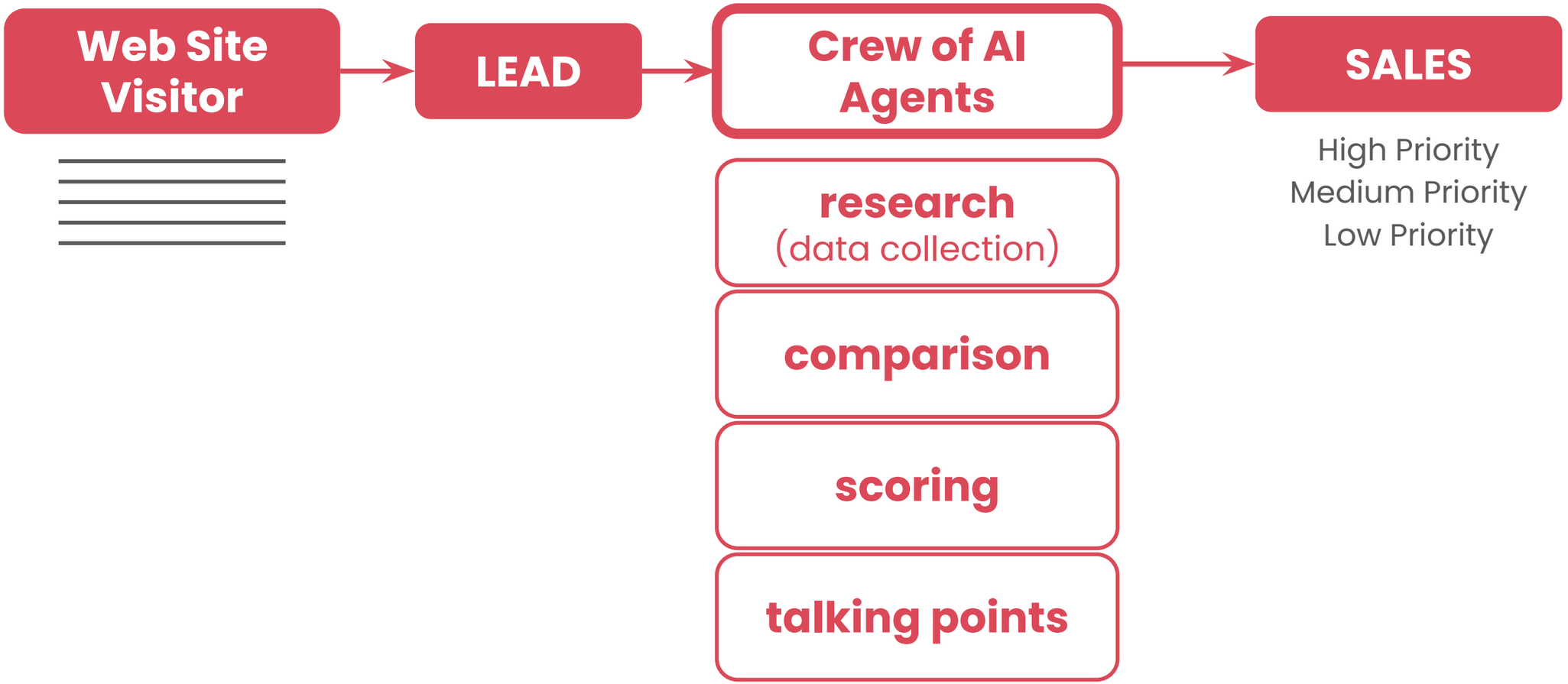
Data Collection and Analysis



Data Collection and Analysis



Data Collection and Analysis



What are AI Agents?

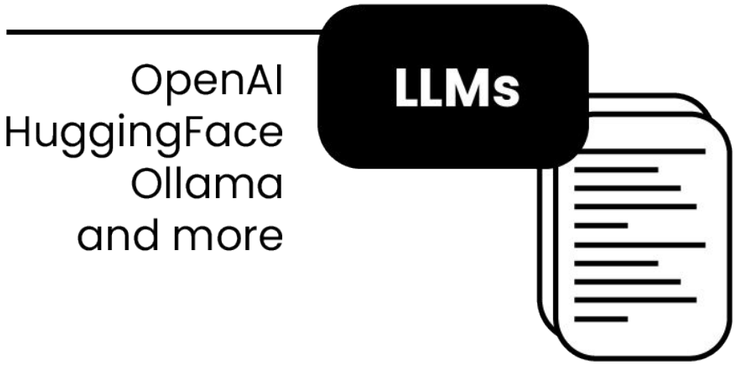
Multi AI Agent Systems with crewAI

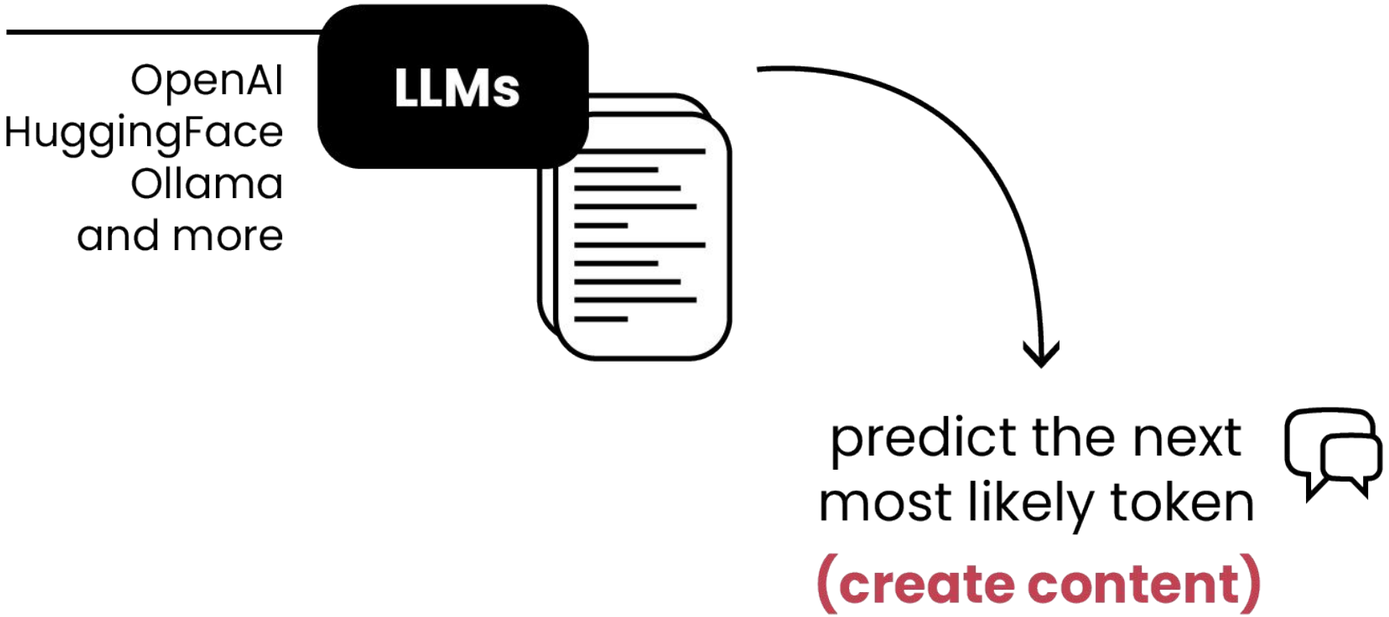
AI Agents





DeepLearning.AI

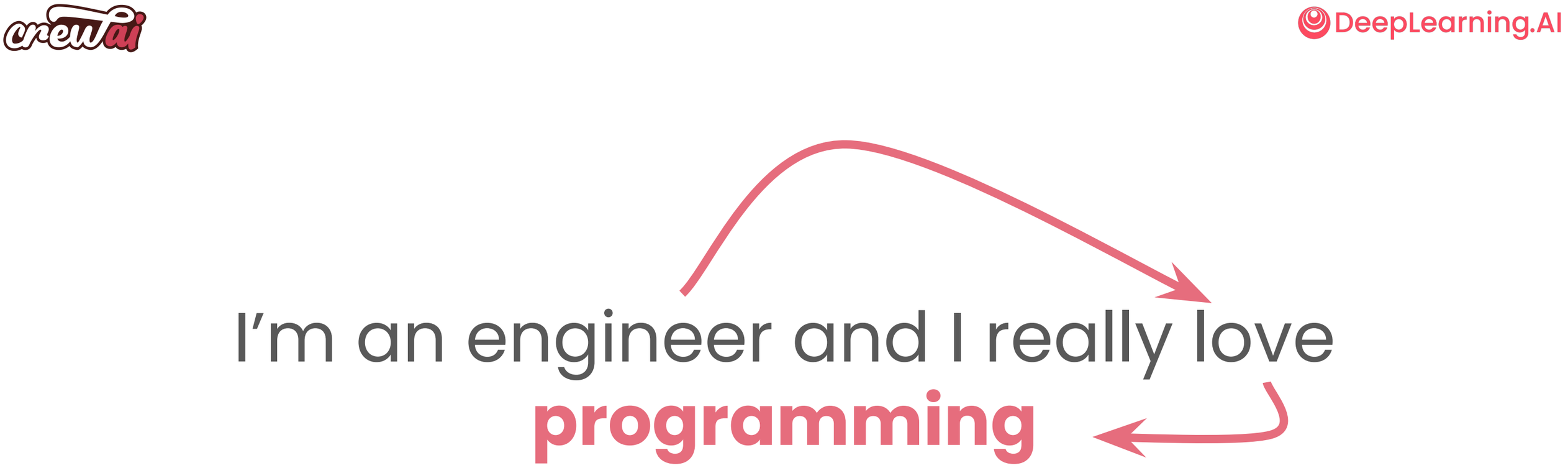

OpenAI
HuggingFace
Ollama
and more
LLMs

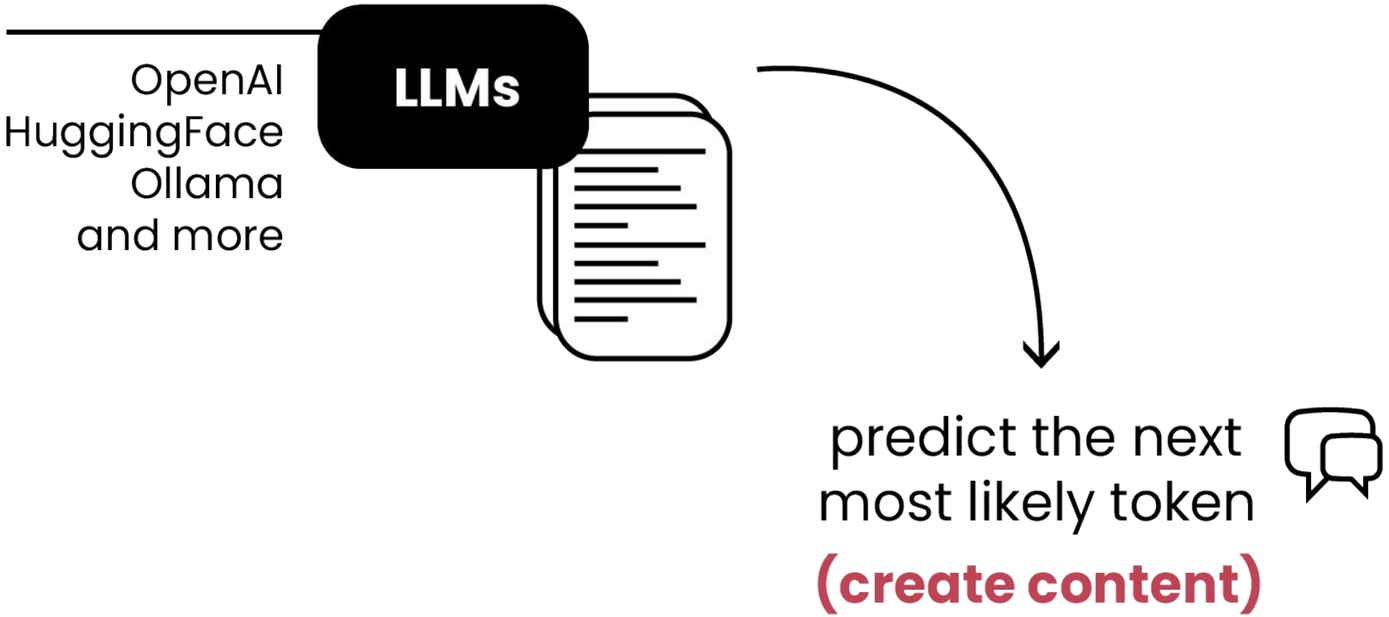

OpenAI
HuggingFace
Ollama
and more
LLMs
predict the next
most likely token
(create content)

I’m an engineer and I really love

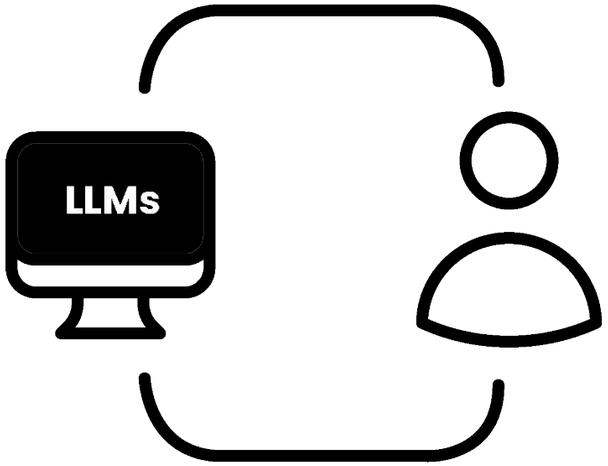
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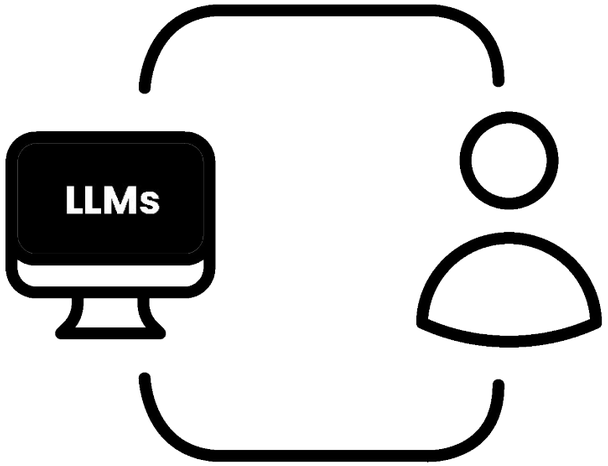

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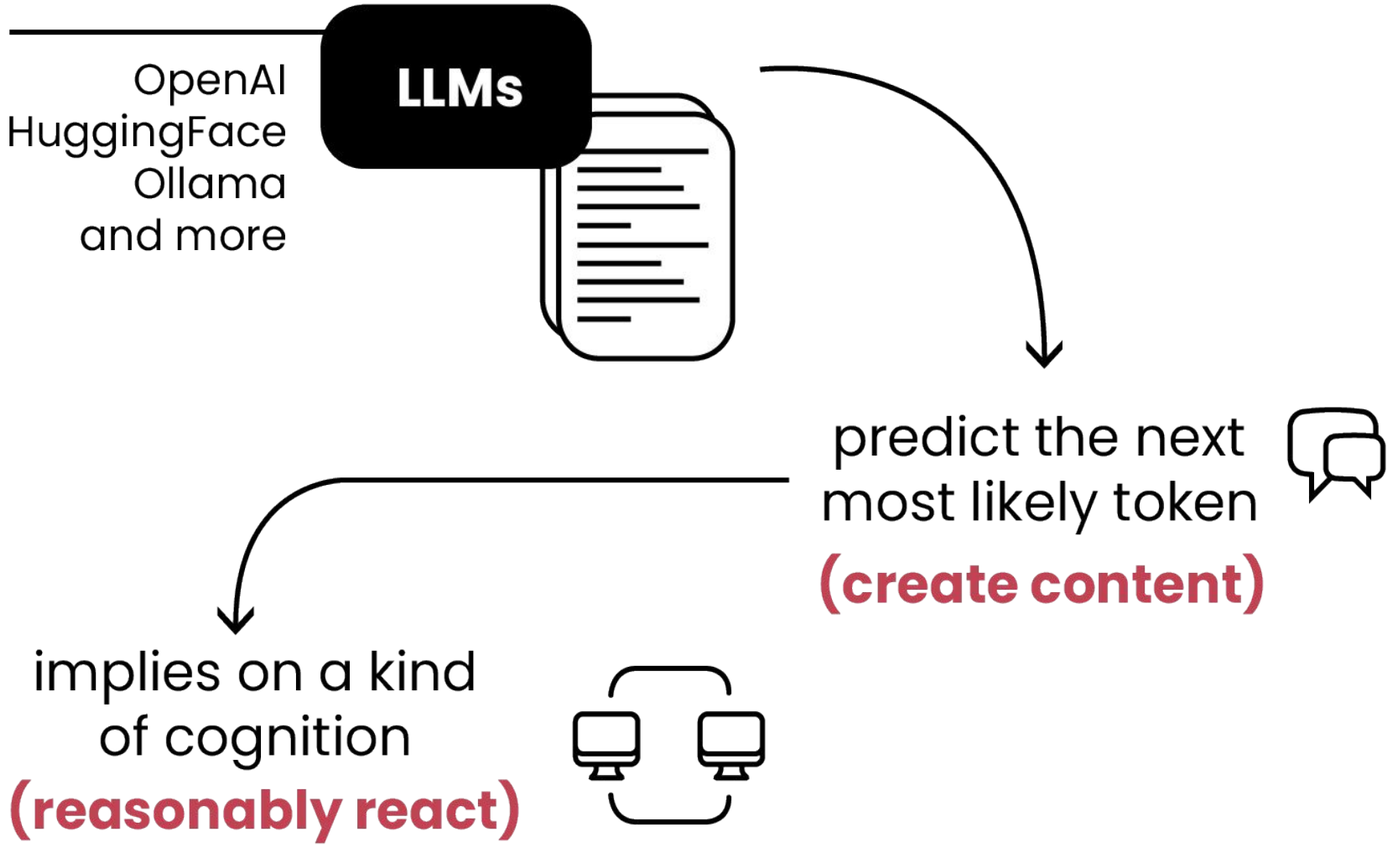
Regular Prompting Experience

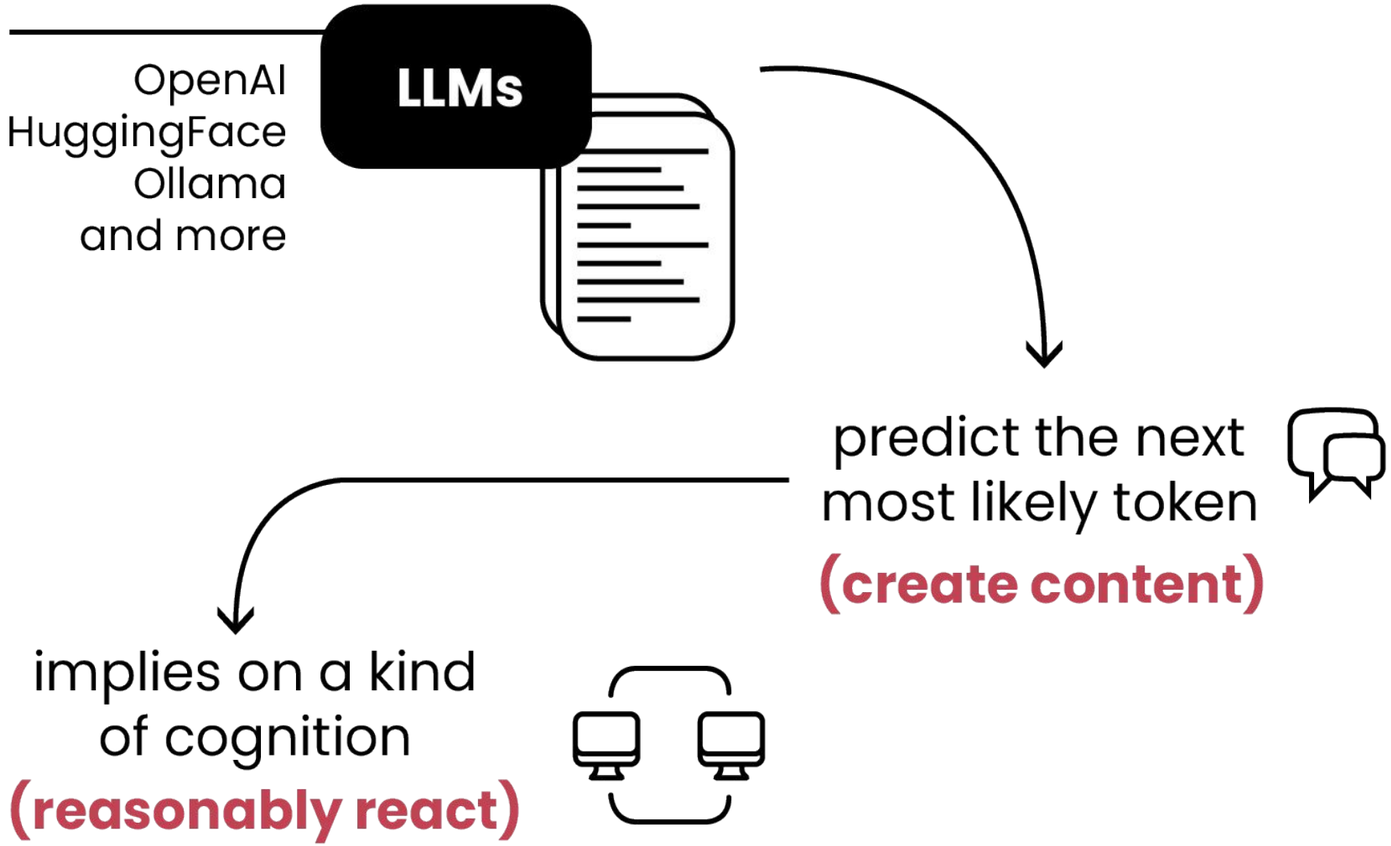


Let’s use chatGPT!

Regular Prompting Experience

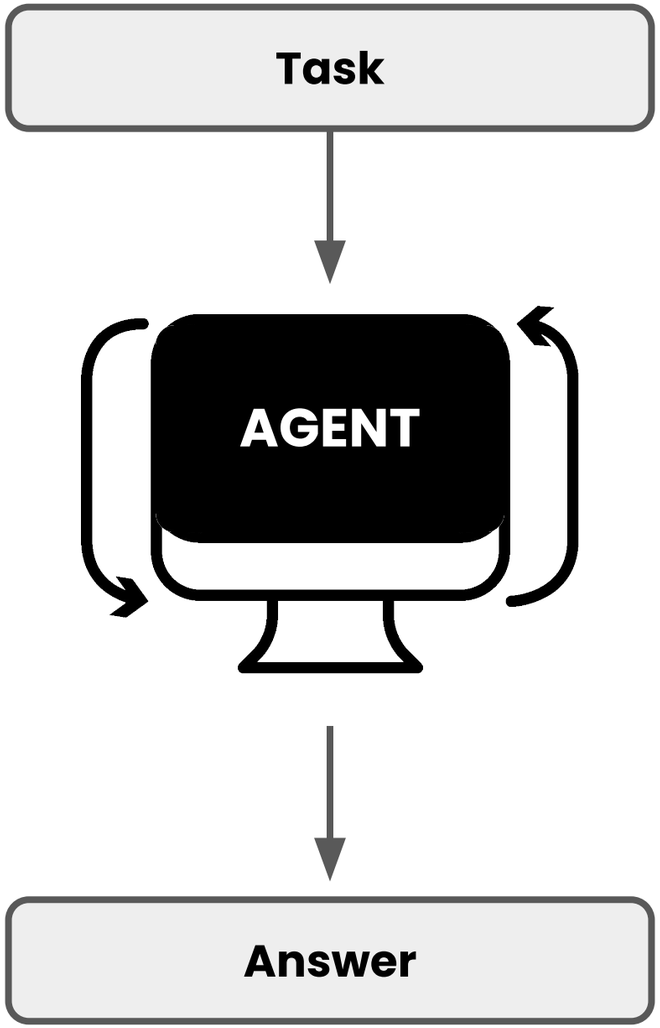


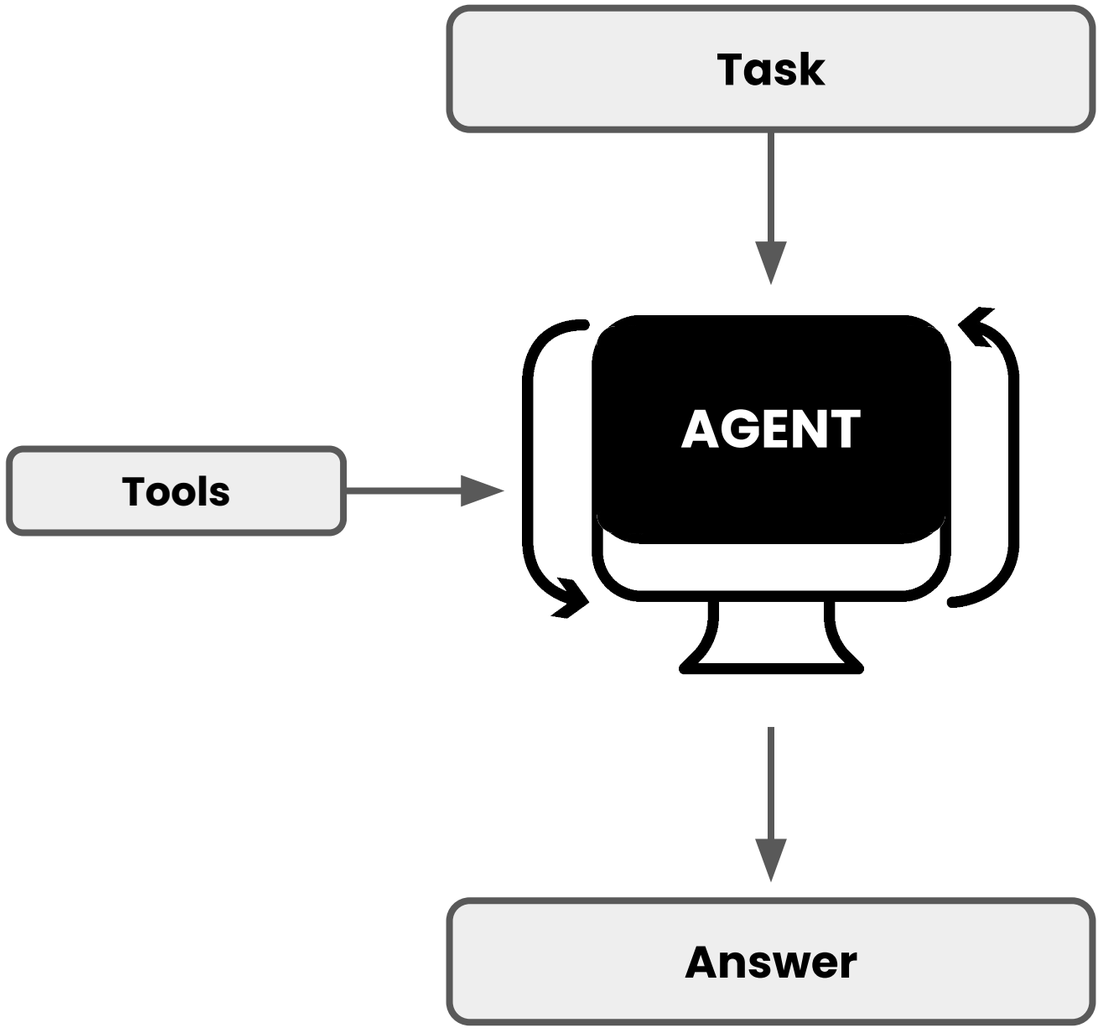

OpenAI
HuggingFace
Ollama
and more
LLMs
predict the next most likely token
(create content)
implies on a kind of cognition
(reasonably react)



An agent is born 🐣

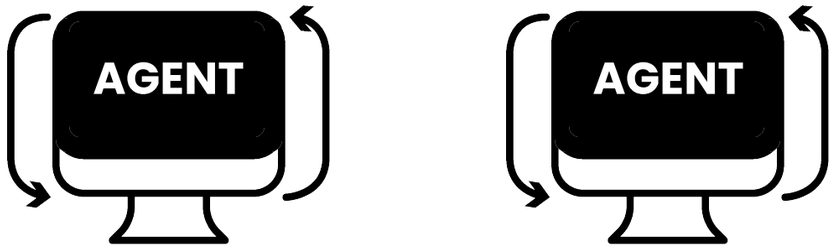

AGENT

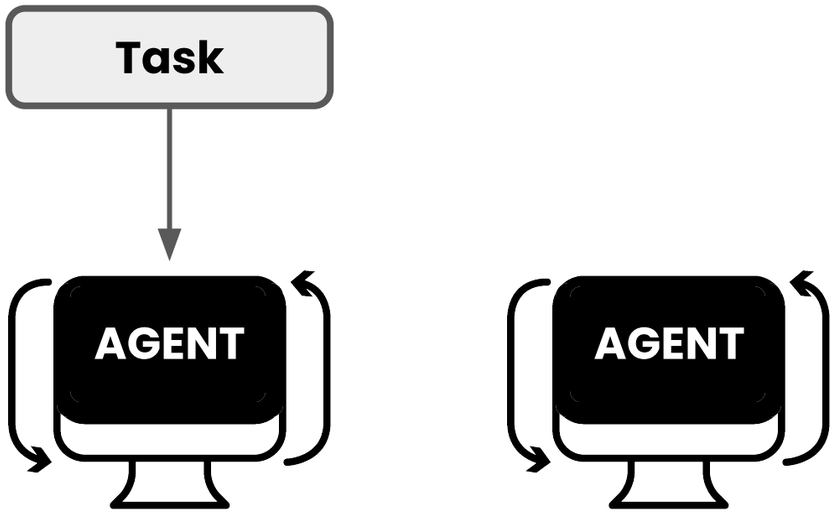

Task
AGENT
Answer

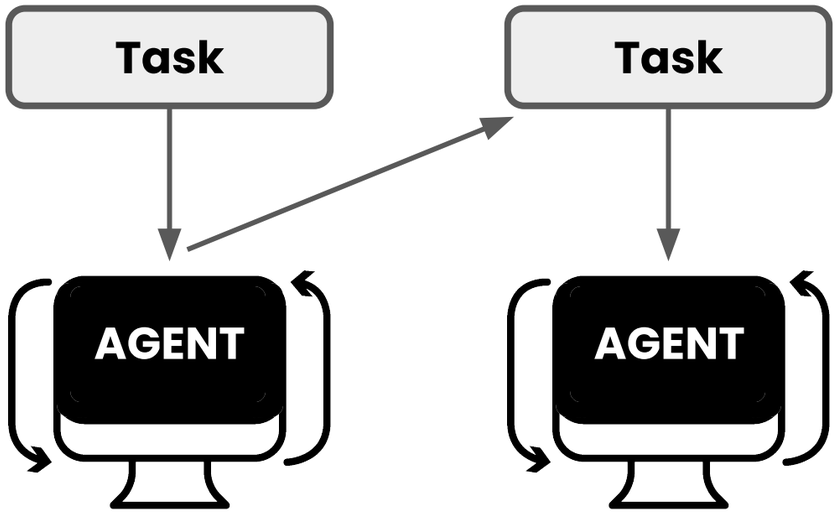

Task
AGENT
Tools
Answer

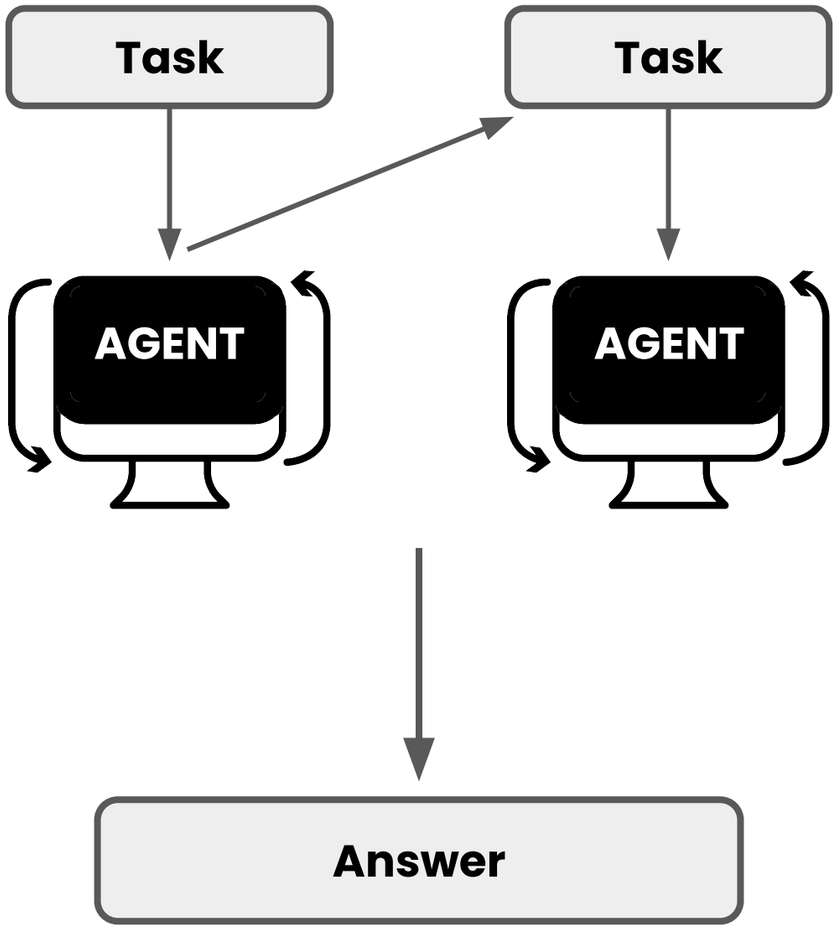
Multi Agent Systems

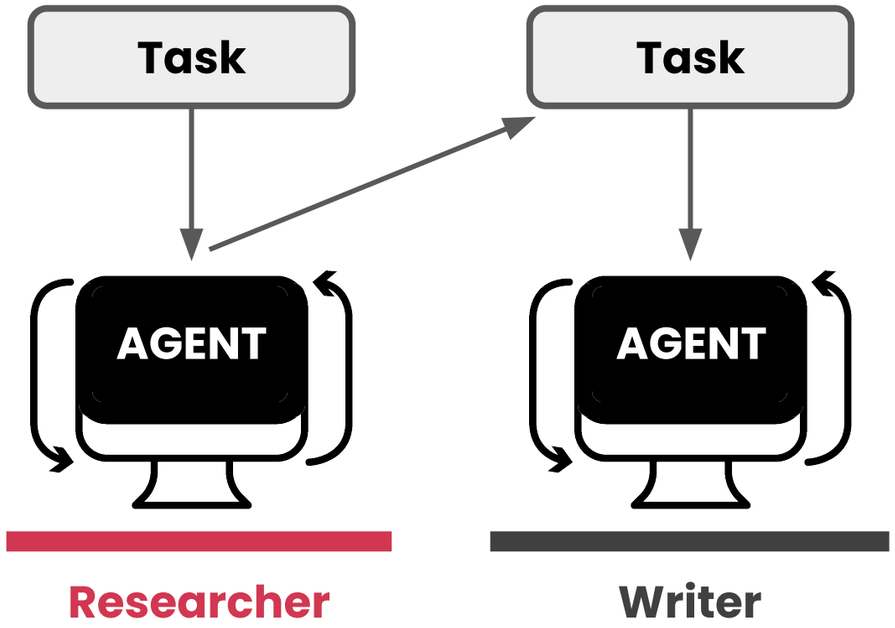

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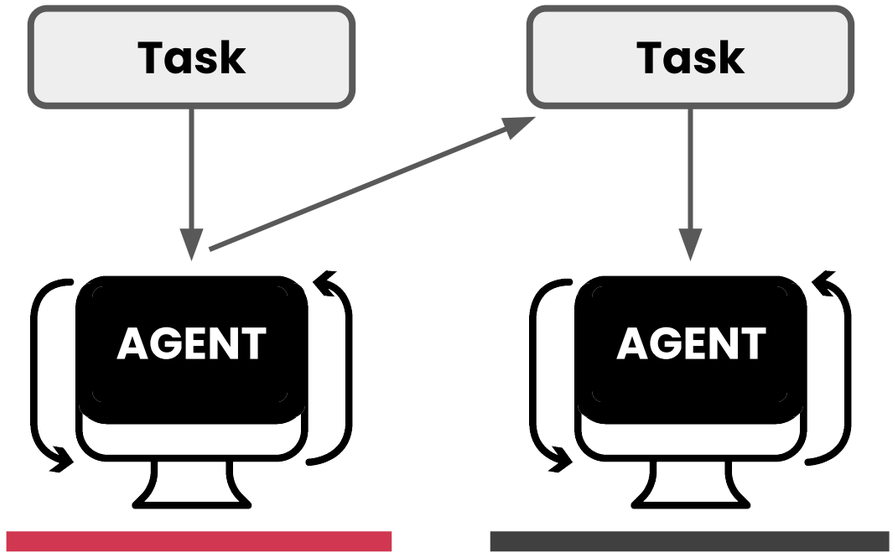

AGENT
AGENT

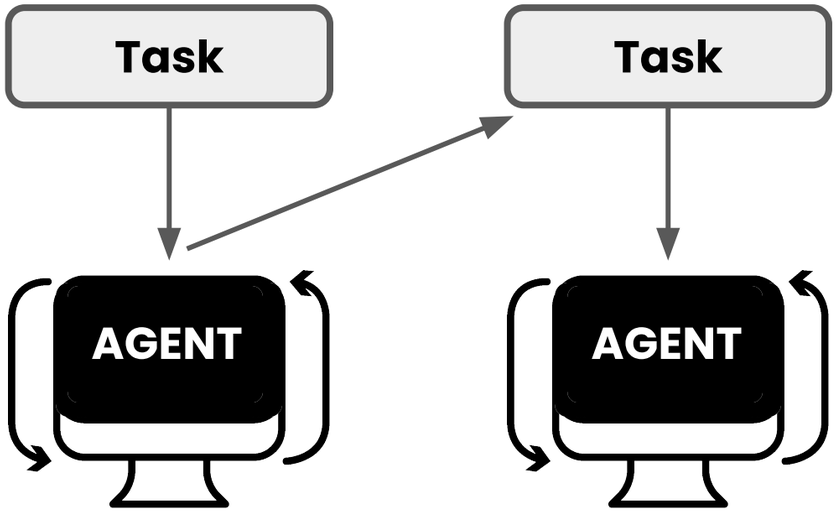

Task
AGENT
AGENT

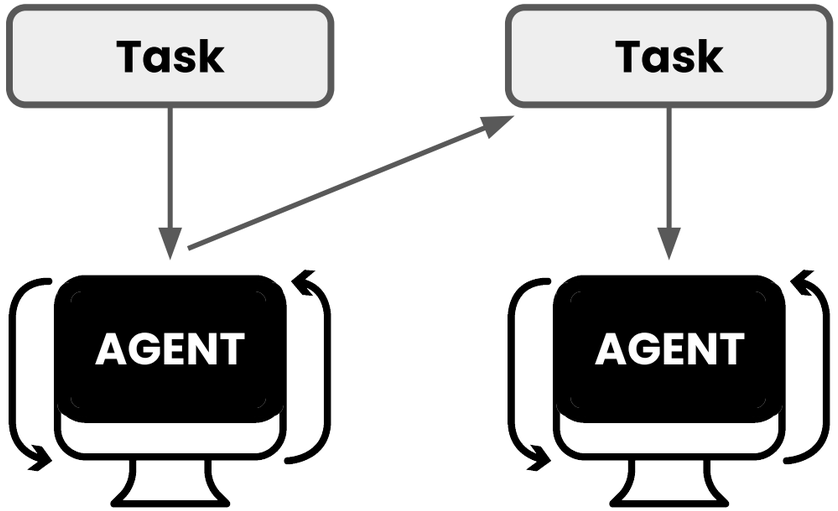

Task
Task
AGENT
AGENT


Task
Task
AGENT
AGENT
Answer


Task
Task
AGENT
AGENT
Researcher
Writer


Task
Task
AGENT
AGENT
Researcher
Llama 3 70B
Writer
GPT-4


Task
Task
AGENT
AGENT


Task
Task
AGENT
AGENT

Framework & Platform

- Breaks all these concepts into simple structures
- Provides a pattern to put these systems together
- Provides tools / skills ready to be used by agents
- Gives you a model to build custom tools for agents
- Offers a platform for bringing these agents into production

What is crewAI?

crewai.com

Initial Building Blocks

Agents

Tasks

Crews

L3A starts here

Multi AI Agent Systems with crewAI

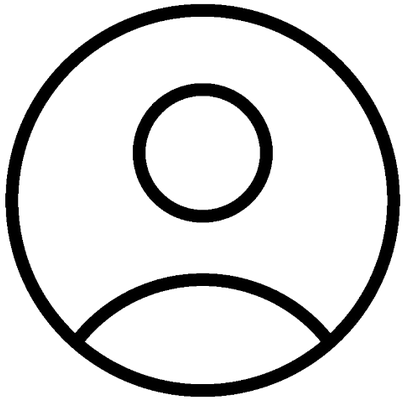
Key elements of AI agents

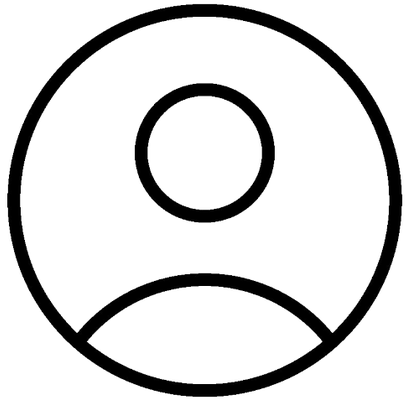




DeepLearning.AI

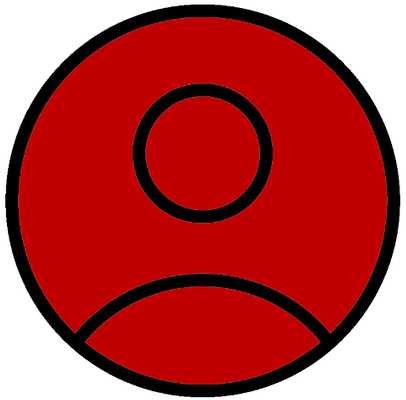
What makes a great Agent?





Role Playing	Focus	Tools
Cooperation	Guardrails	Memory

Role Playing



Role Playing



You

give me an analysis on tesla stock

Role Playing

You

give me an analysis on tesla stock

You

you are a FINRA approved financial analyst.

give me an analysis on tesla stock

Role Playing



You

give me an analysis on tesla stock



You

you are a FINRA approved financial analyst.

give me an analysis on tesla stock

- Roles
- Goals
- Backstory

ChatGPT

Tesla's stock analysis involves looking at various factors, including financial performance, industry trends, technological advancements, competition, and market sentiment. Here's a brief overview:

1. **Financial Performance**: Tesla's financial performance has been a mix of impressive revenue growth and periodic profitability challenges. Investors often scrutinize metrics like revenue, earnings per share (EPS), and gross margins. Quarterly earnings reports, as well as annual filings, provide insights into the company's financial health.

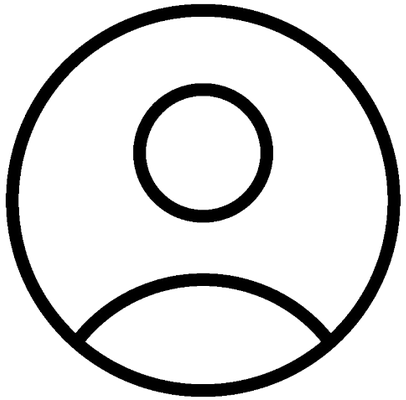
ChatGPT

As a financial analyst, I'd be happy to provide an analysis of Tesla's stock. Please note that this information is for educational purposes only and should not be considered as financial advice.

Company Overview:

Tesla, Inc. (NASDAQ: TSLA) is a leading electric vehicle (EV) manufacturer, energy storage company, and solar panel manufacturer. Founded in 2003, Tesla has become synonymous with innovation in the automotive industry, particularly in the realm of sustainable transportation.

Recent Performance:

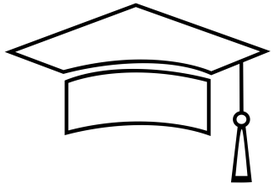


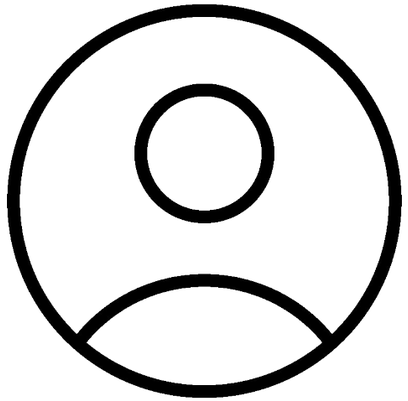
Role Playing	Focus	Tools
Cooperation	Guardrails	Memory

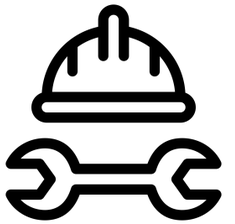
Focus







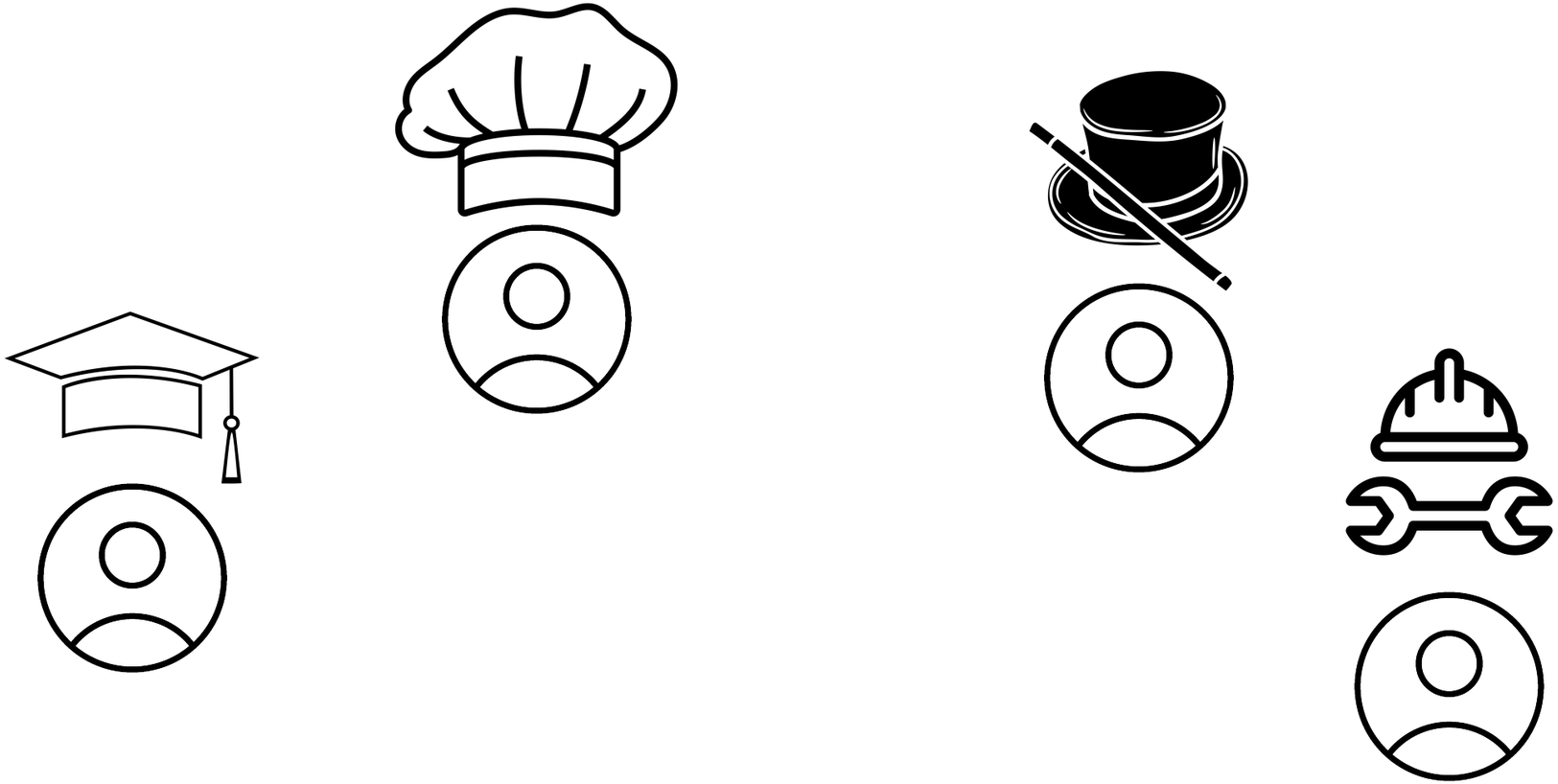


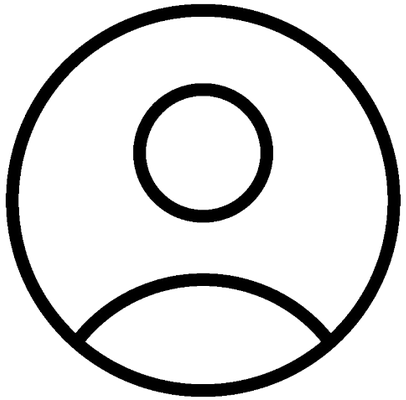


crewai

DeepLearning.AI

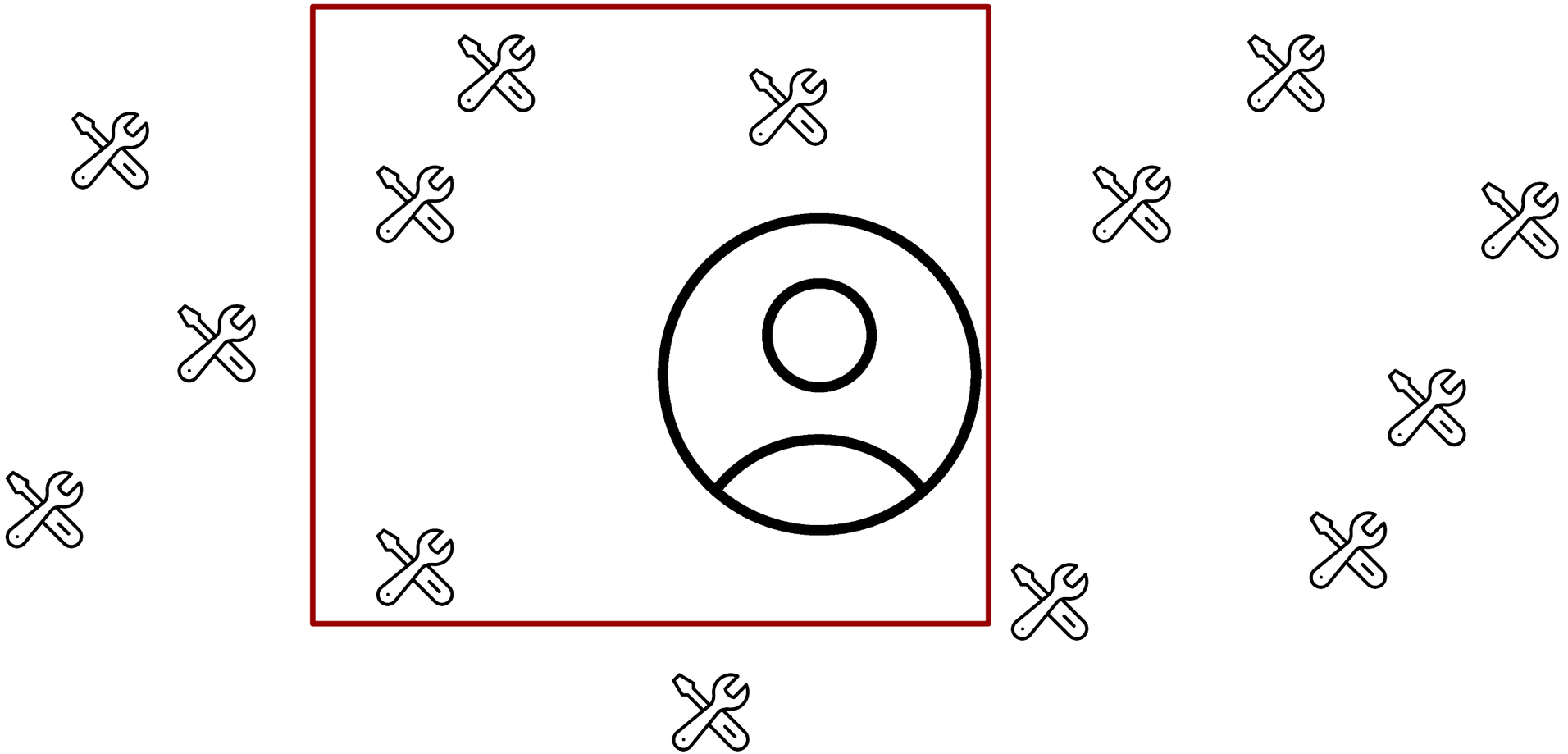
Focus



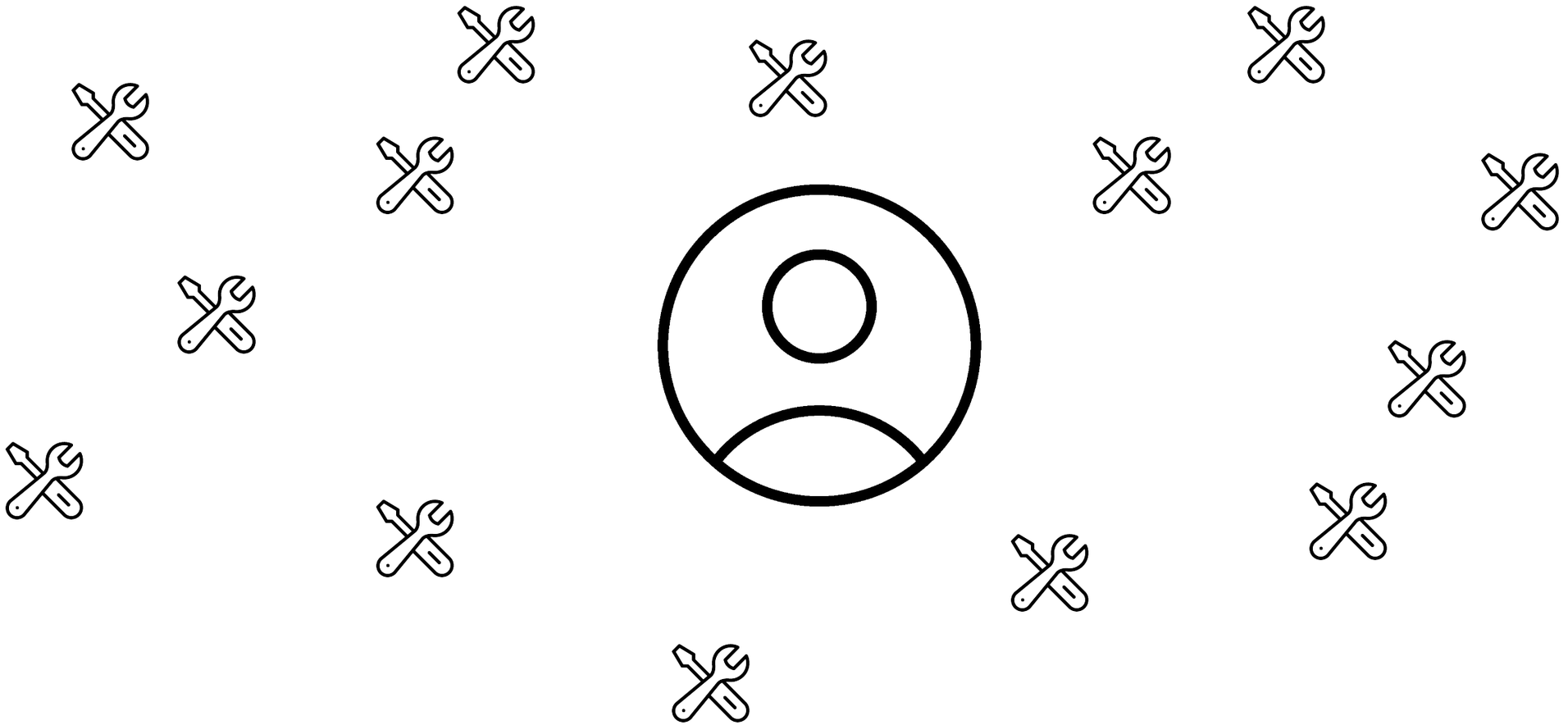


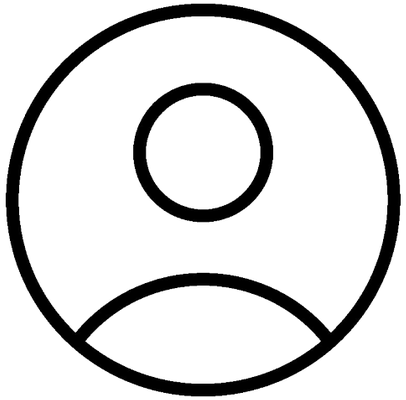
Role Playing	Focus	Tools
Cooperation	Guardrails	Memory

Tools



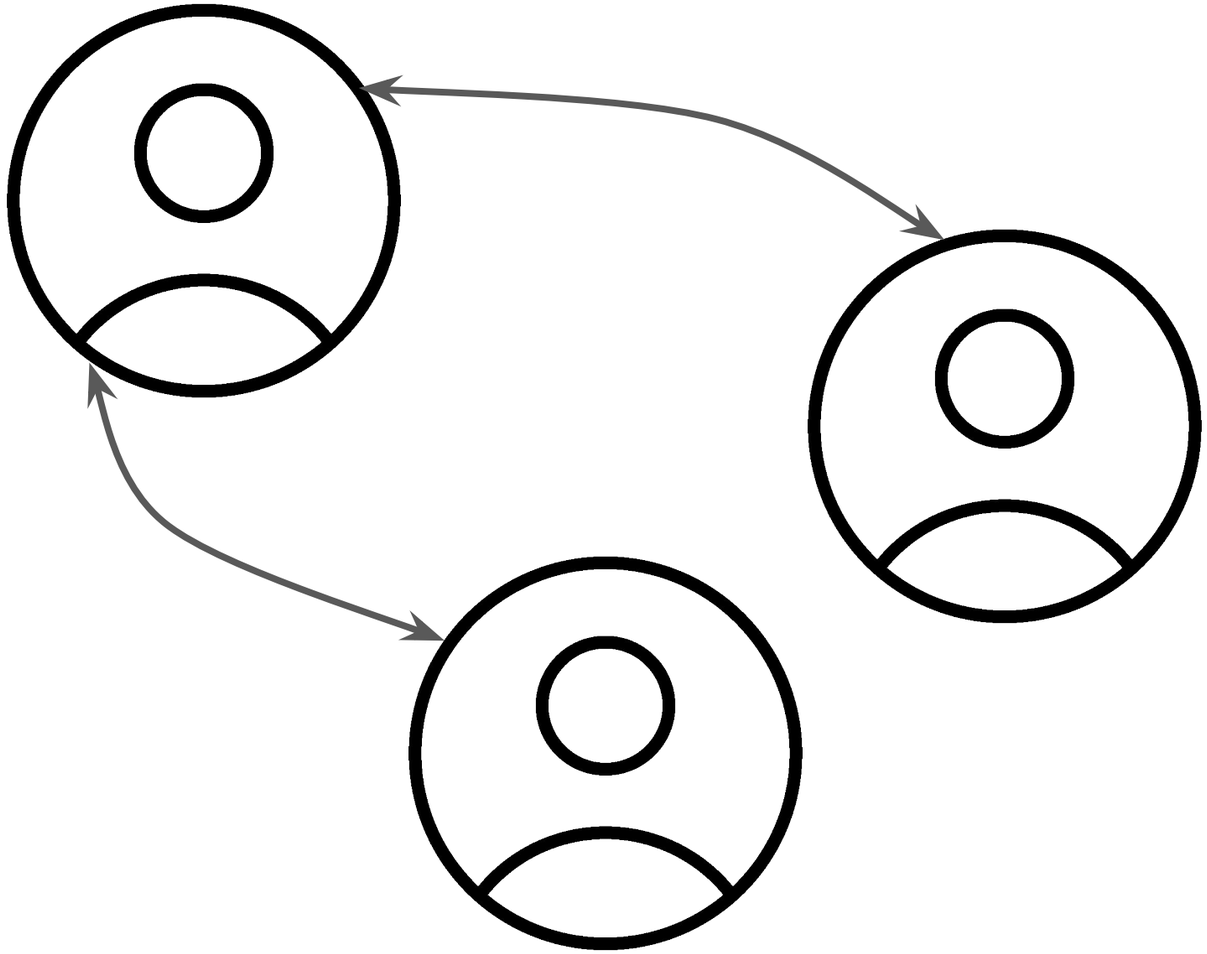
Tools



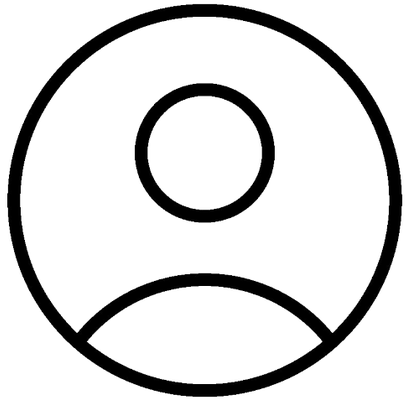


Role Playing	Focus	Tools
Cooperation	Guardrails	Memory

Collaboration

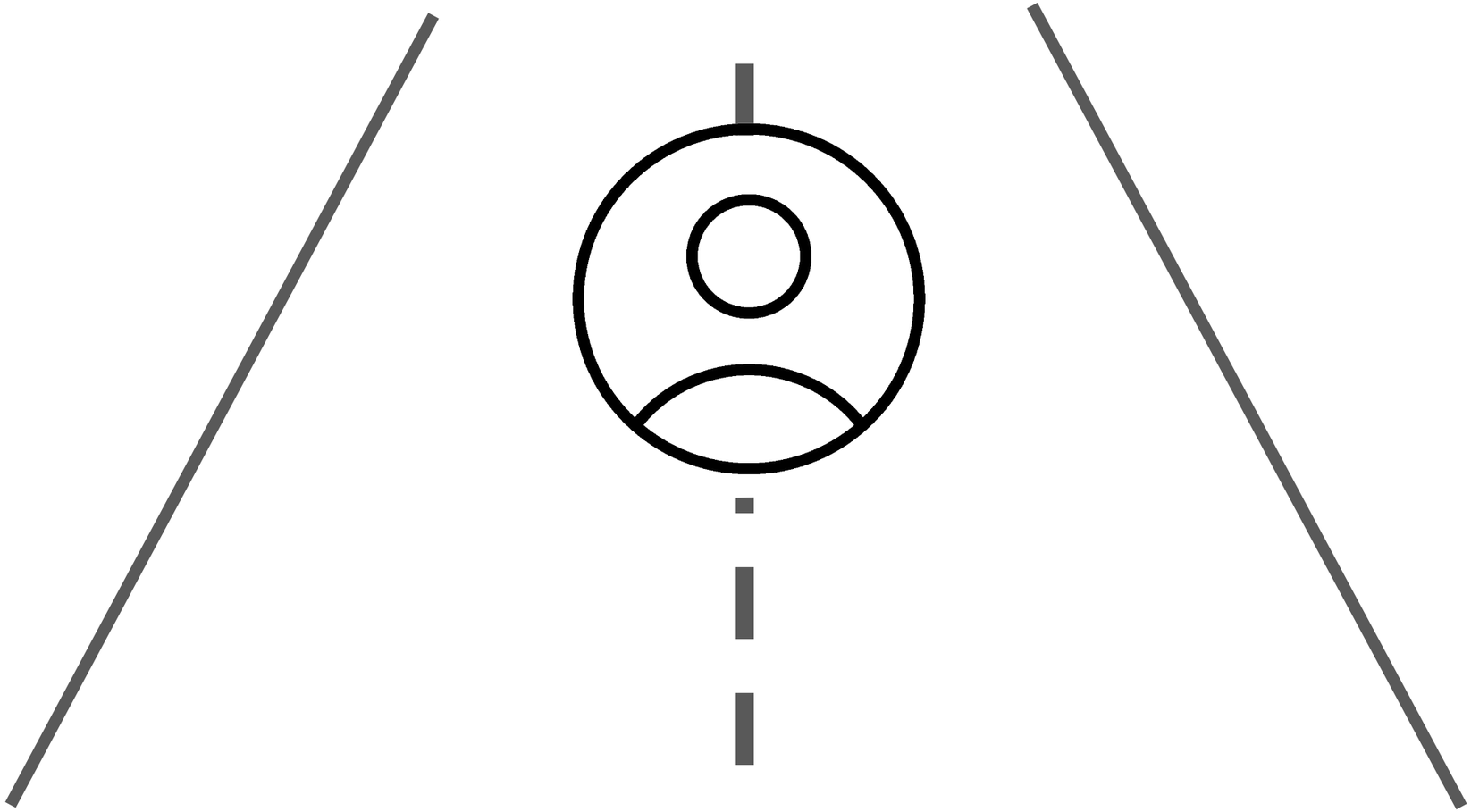


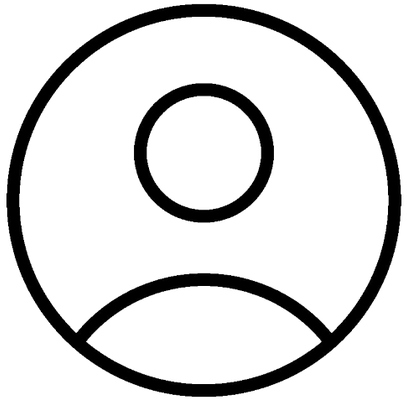
- Take feedback
- Delegate tasks



Role Playing	Focus	Tools
Cooperation	Guardrails	Memory

Guardrails





Role Playing

Focus

Tools

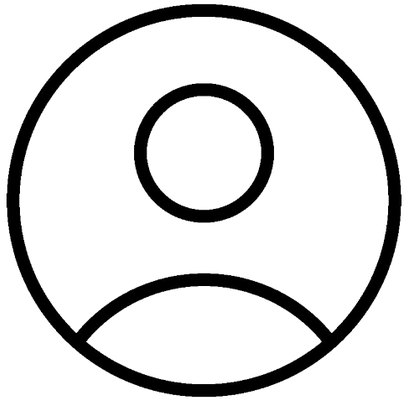
Cooperation

Guardrails

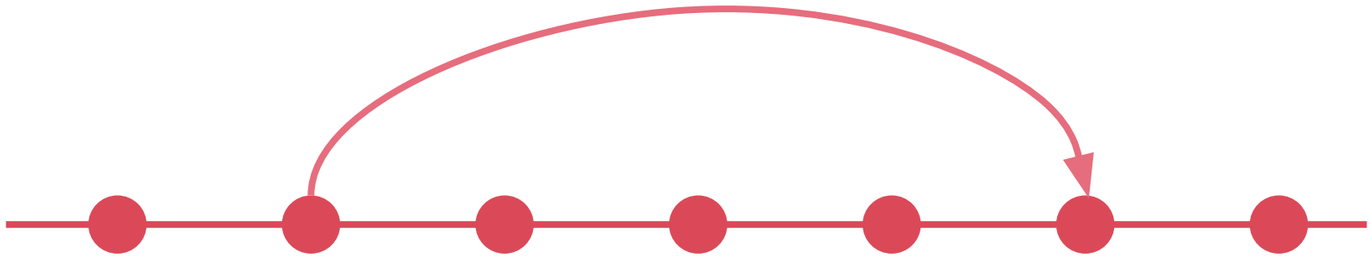
Memory

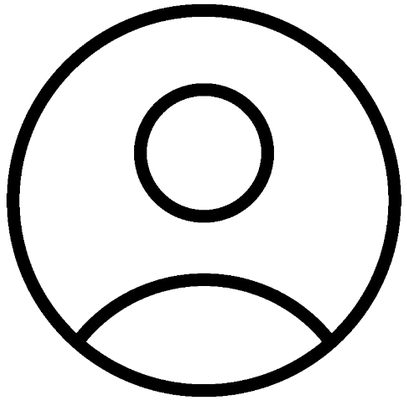
Memory



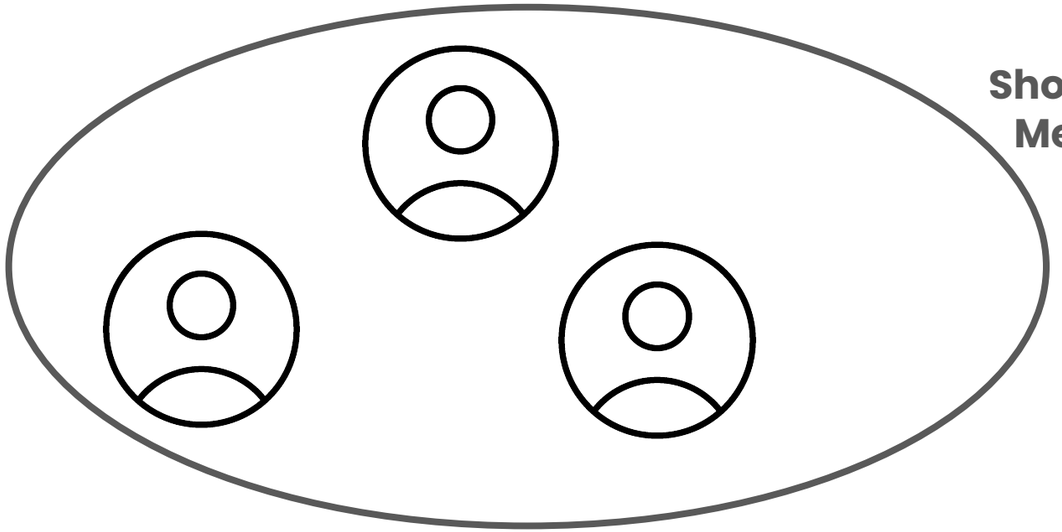


Memory

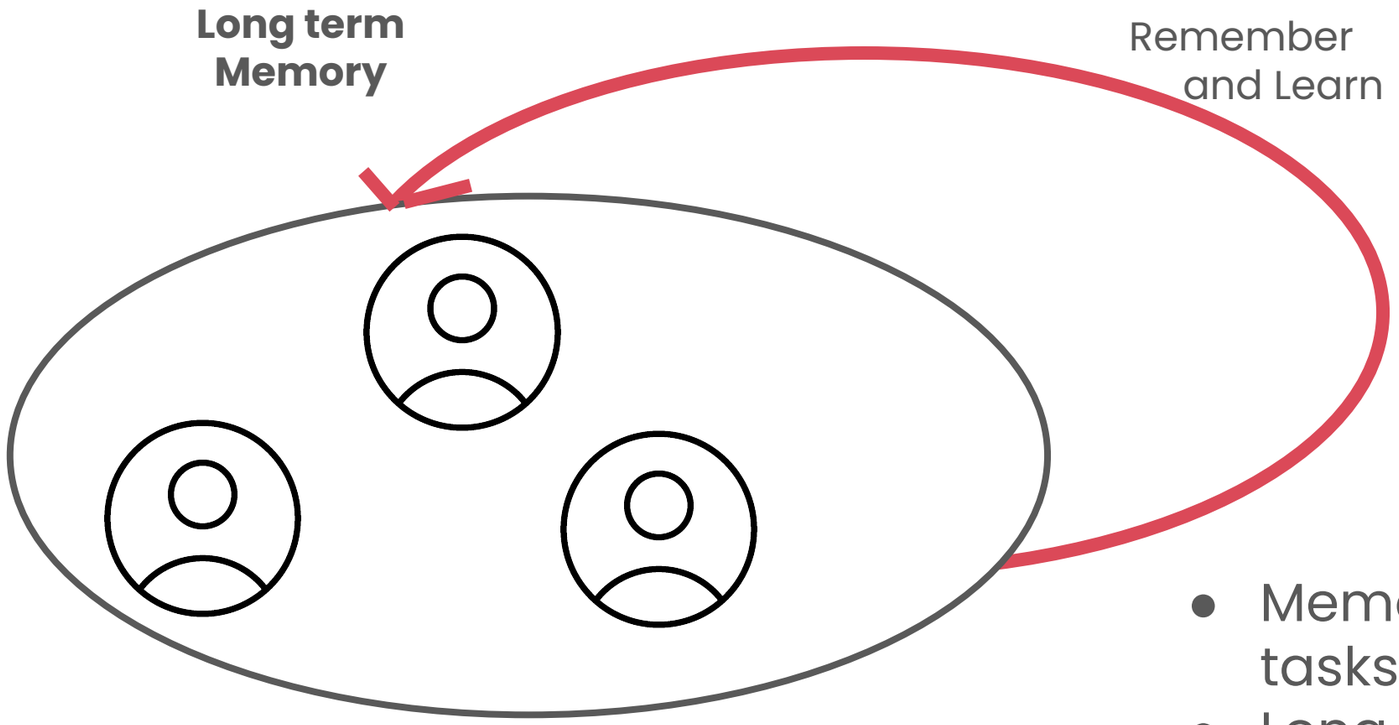




1. Long-term memory
2. Short-term memory
3. Entity memory



- Memory during execution of a task.
- Share knowledge, activities, and learnings with other agents.
- Share intermediate information even before providing "task completion" output.



- Memory stored after execution of current tasks.
- Long term memory can be used in any future tasks.
- Leads to "self-improving" agents.

Entity Memory

Pretend news article (by ChatGPT)

Renewable energy corporations such as Tesla, led by CEO Elon Musk, and Orsted, based in Denmark, are at the forefront of the transition to sustainable energy. Meanwhile, tech giants like Google, under the leadership of Sundar Pichai, and Microsoft, with CEO Satya Nadella at the helm, are investing heavily in renewable energy projects and carbon offset programs to achieve carbon neutrality. Notably, countries like Germany, with its ambitious Energiewende policy, and China, the world's largest emitter of greenhouse gases, are implementing aggressive strategies to reduce emissions and accelerate the adoption of renewable energy. Additionally, startups like Beyond Meat, founded by Ethan Brown, and Impossible Foods, led by CEO Pat Brown, are disrupting the food industry with plant-based alternatives, significantly reducing the environmental impact of food production.

Entities

- persons,
- organizations,
- locations

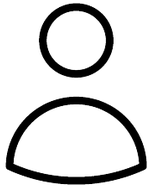
Multi AI Agent Systems with crewAI

Mental framework for agent creation

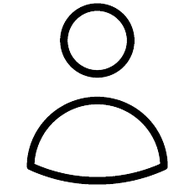




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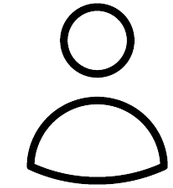
Manager



Manager

What is the Goal

What is the Process

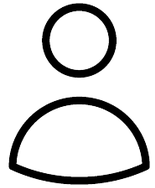


Manager

What is the Goal

What is the Process

What kind of people would I need to hire to get this done?



What is the Goal

What is the Process

What kind of people would I need to hire to get this done?



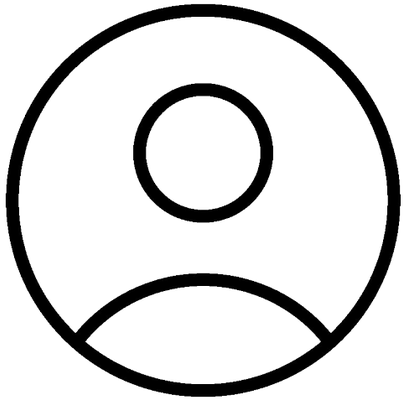



- **"Okay" Agent roles**
  - Researcher
  - Writer
  - Financial Analyst

- **Okay**
  - Researcher
  - Writer
  - Financial Analyst

- **Better**
  - HR Research Specialist
  - Senior copywriter
  - FINRA Approved Analyst

Let’s recap!



Role Playing

Focus

Tools

Cooperation

Guardrails

Memory



- Agents "self improve" using memory
- Guardrails prevents agents from going into "rabbit holes" (unproductive, repetitive loops)
- Agents always attempt to get to an answer (avoid iterating indefinitely)
- Focus
  - narrowly defined task
  - specific agent roles and objectives
  - limited set of tools assigned to one agent

Lesson 4A

Multi AI Agent Systems with crewAI

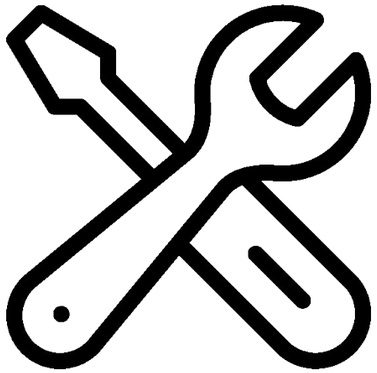
Key elements of agent tools



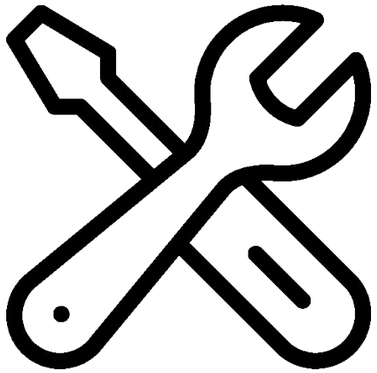


DeepLearning.AI

What makes a great Tool?



What makes a great Tool?

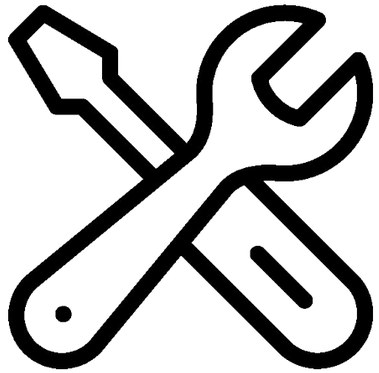


Versatile

Fault-tolerant

Caching

What makes a great Tool?



- Versatile
- Fault-tolerant
- Caching

Versatile

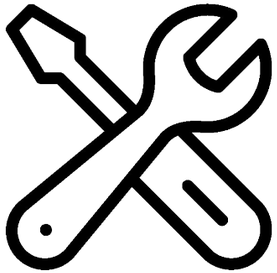
Agent



fuzzy inputs

- text instructions Mix of words, numbers

tool



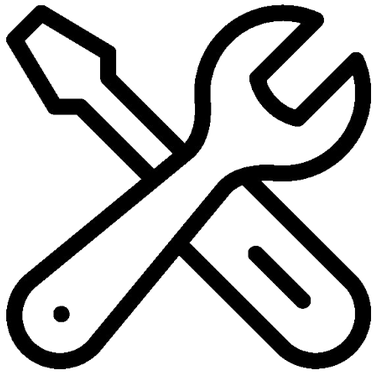
strongly typed output

- JSON (key-value pairs)
- list of strings, categories
- numbers

External world
(traditional software systems)



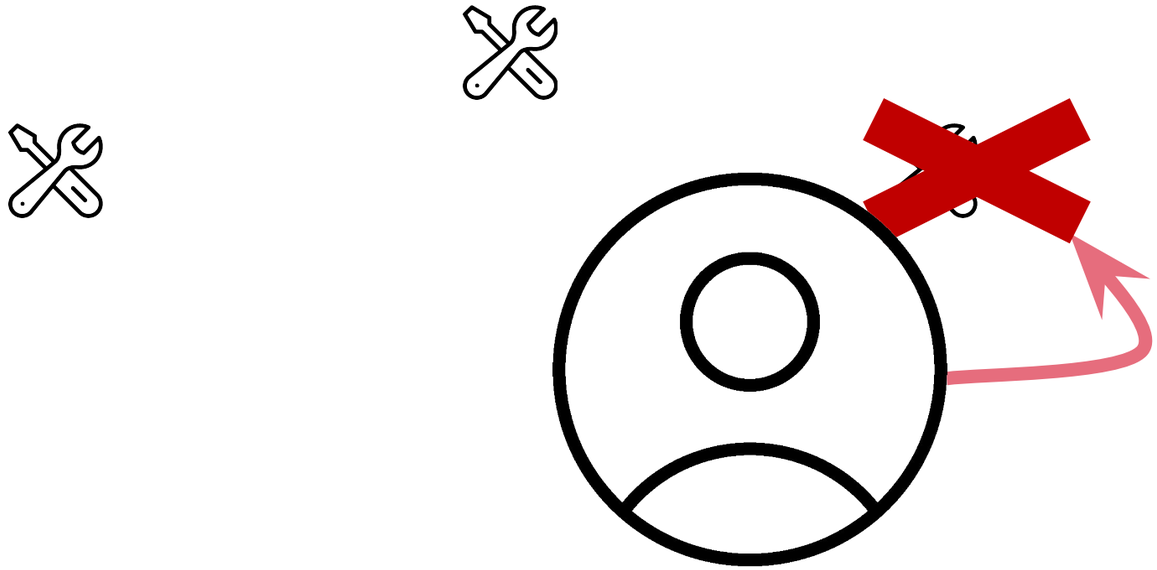
What makes a great Tool?



Versatile

Fault-tolerant

Caching



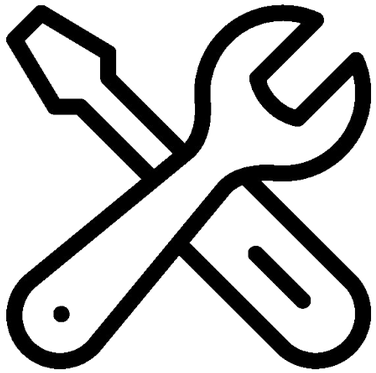
What happens when an exception/error occurs?

Option 1: stop execution

Option 2: Fail gracefully (and try again)

- send error message back to agent
- Ask agent to retry given the error message

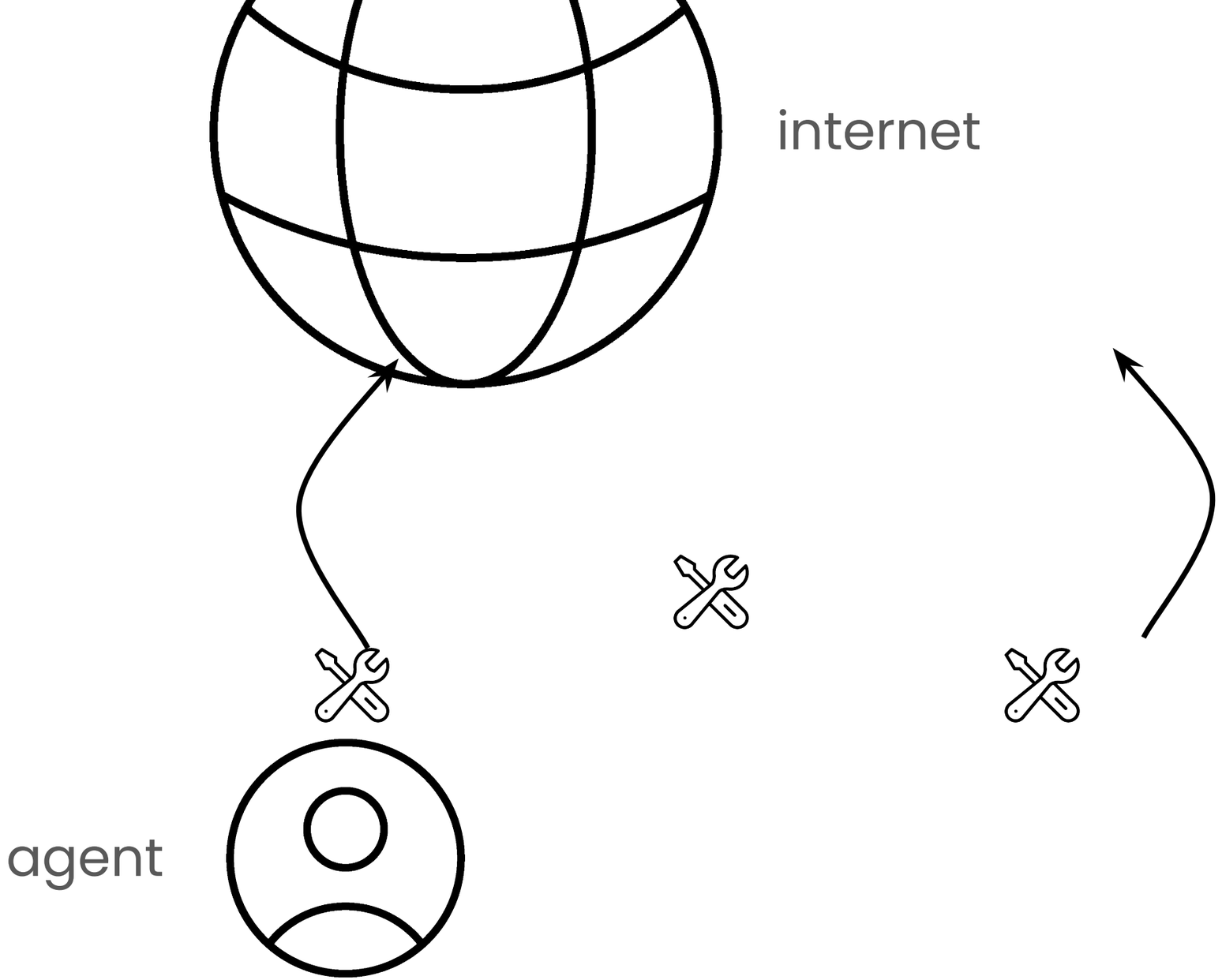
What makes a great Tool?

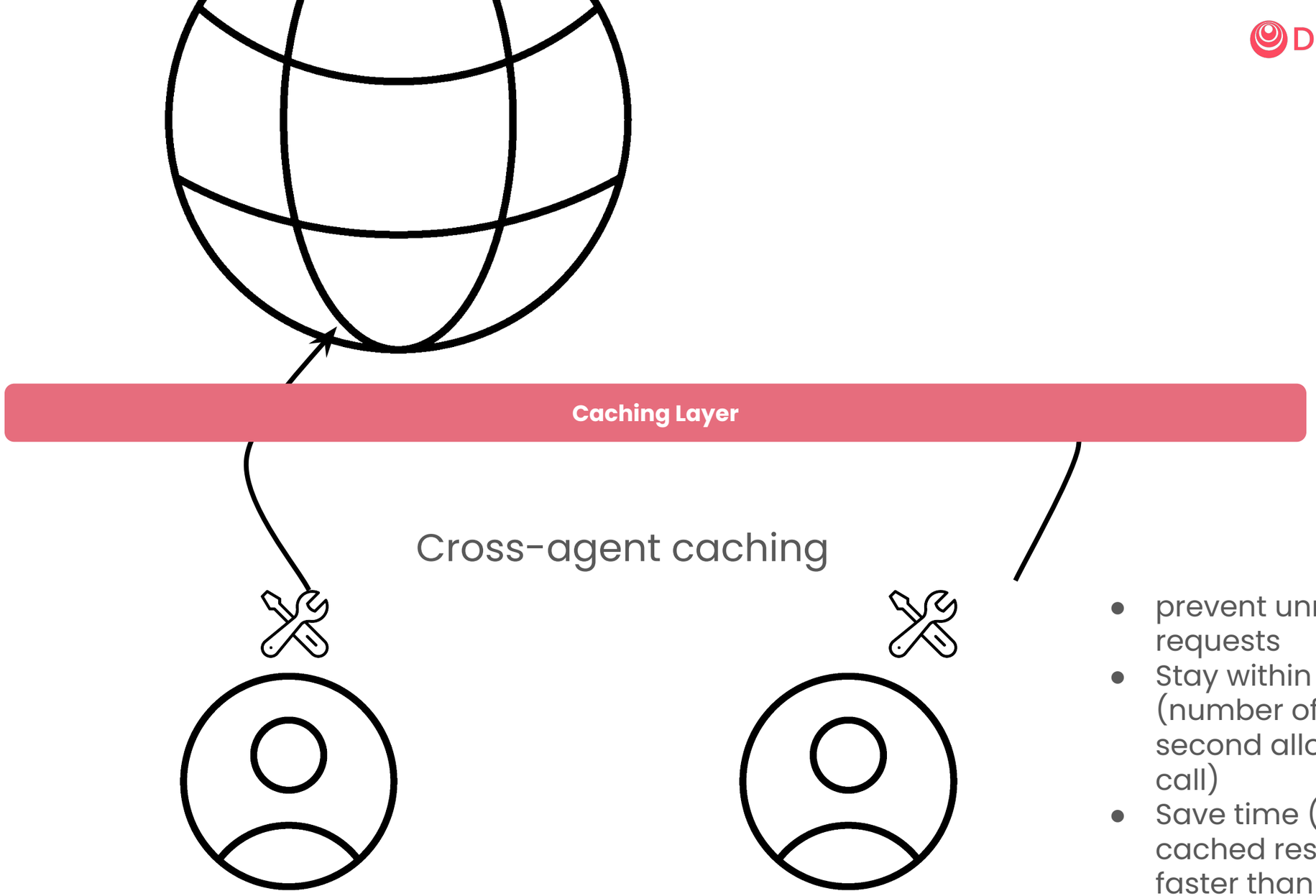


Versatile

Fault-tolerant

Caching


internet
agent



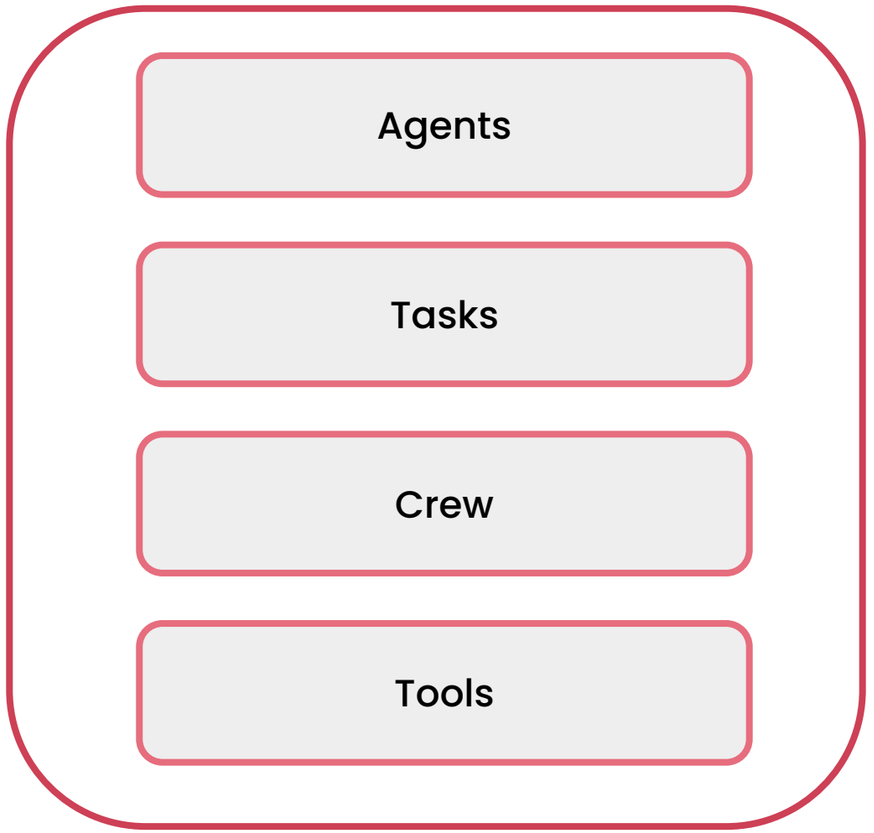
Cross-agent caching

- prevent unnecessary requests
- Stay within rate limits (number of requests per second allowed for an API call)
- Save time (retrieving cached results is much faster than an API call)

Examples of tools

- Search the internet
- Scrape a website
- Connect to a database
- Call an API
- Send notifications

and many more


Agents
Tasks
Crew
Tools

Google Serper Search
Scrape Website
Directory Read
File Read
Selenium Scraper
Directory RAG Search
PDF RAG Search
TXT RAG Search
CSV RAG Search
XML RAG Search
JSON RAG Search
Docx Rag Search
MDX RAG Search
PG RAG Search
Website RAG Search
Github RAG Search
Code Docs RAG Search
Youtube Video RAG Youtube
Channel RAG Search

crewAI
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Core Concepts
How to Guides
Tools Docs
Google Serper Search
Scrape Website
Directory Read
File Read
Selenium Scraper
Directory RAG Search
PDF RAG Search
TXT RAG Search
CSV RAG Search
XML RAG Search
JSON RAG Search
Docx Rag Search
MDX RAG Search
PG RAG Search
Website RAG Search
Github RAG Search
Code Docs RAG Search
Youtube Video RAG Search
Youtube Channel RAG Search
Examples
Telemetry



crewAI Documentation

Cutting-edge framework for orchestrating role-playing, autonomous AI agents. empowers agents to work together seamlessly, tackling complex tasks.

Core Concepts

- Agents
- Tasks
- Tools
- Processes
- Crews
- Memory

How-To Guides

- Getting Started
- Create Custom Tools
- Using Sequential Process
- Using Hierarchical Process
- Connecting to LLMs
- Customizing Agents
- Human Input on Execution
- Agent Observability using AgentOps

crewAI
Home
Core Concepts
How to Guides
Tools Docs
Google Serper Search
Scrape Website
Directory Read
File Read
Selenium Scraper
Directory RAG Search
PDF RAG Search
TXT RAG Search
CSV RAG Search
XML RAG Search
JSON RAG Search
Docx Rag Search
MDX RAG Search
PG RAG Search
Website RAG Search
Github RAG Search
Code Docs RAG Search
Youtube Video RAG Search
Youtube Channel RAG Search
Examples
Telemetry



crewAI Documentation

Cutting-edge framework for orchestrating role-playing, autonomous AI agents. empowers agents to work together seamlessly, tackling complex tasks.

Core Concepts

- Agents
- Tasks
- Tools
- Processes
- Crews
- Memory

How-To Guides

- Getting Started
- Create Custom Tools
- Using Sequential Process
- Using Hierarchical Process
- Connecting to LLMs
- Customizing Agents
- Human Input on Execution
- Agent Observability using AgentOps


crewai
+
LangChain

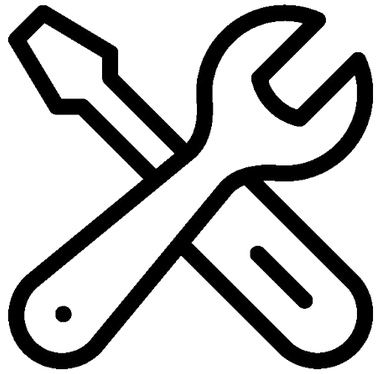
Multi AI Agent Systems with crewAI

Recap of tools





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Versatile

Fault-tolerant

Caching

Lesson 5A

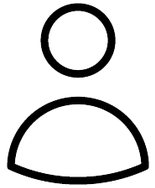
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Key elements of well defined tasks





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Manager

What is the Goal

What is the Process

What kind of individuals would I need to hire to get this done?






Which processes and tasks do I expect the individuals on my team to do?

Task

Task

Task

Task

Task



Which processes and tasks do I expect the individuals on my team to do?

Task

- Clear description of the task
- Set a clear and concise expectation
- Set context
- Set a callback
- Override Agent tools with specific task tools
- Force human input before end of task
- Execute asynchronously
- Output as Pydantic
- Output as JSON
- Output as file
- Run in parallel

Lesson 5C

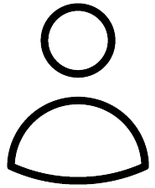
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Recap of tasks





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Manager

What is the Goal

What is the Process

What kind of individuals would I need to hire to get this done?






Which processes and tasks do I expect the individuals on my team to do?

Task

Task

Task

Task

Task

Lesson 6A

Multi AI Agent Systems with crewAI

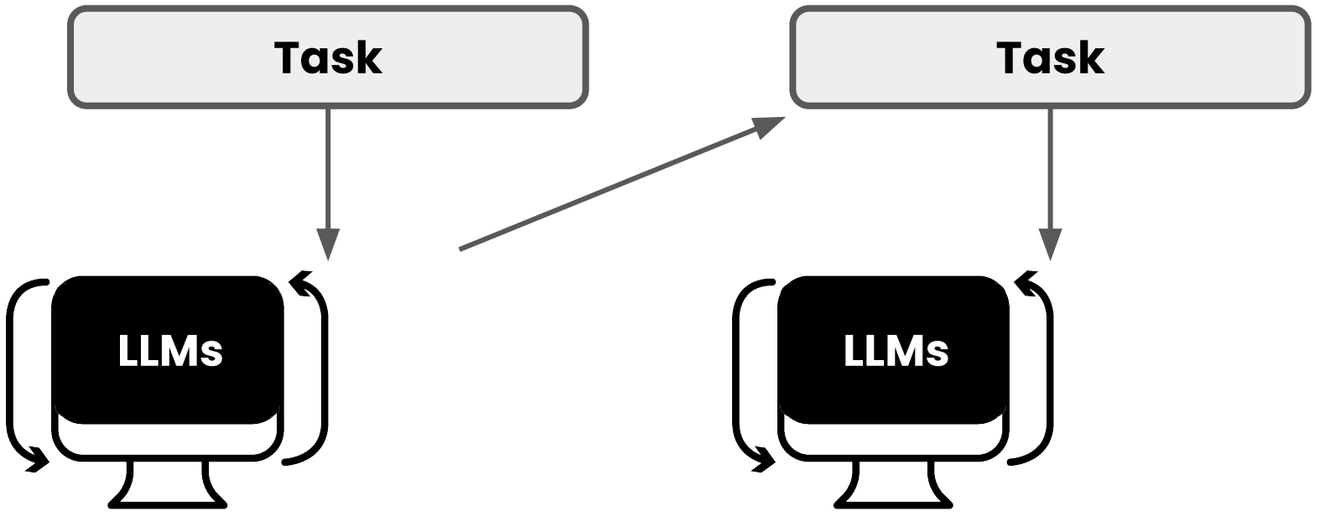
Multi agent collaboration

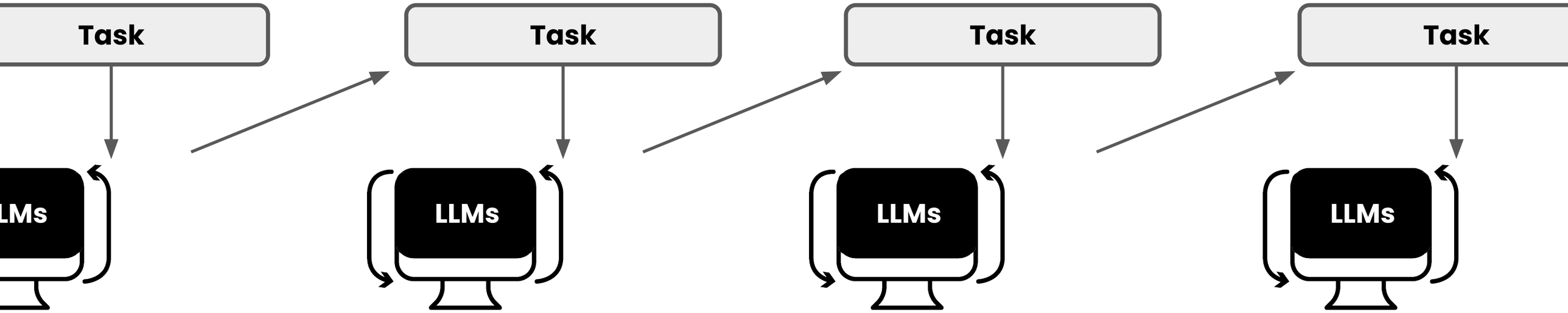


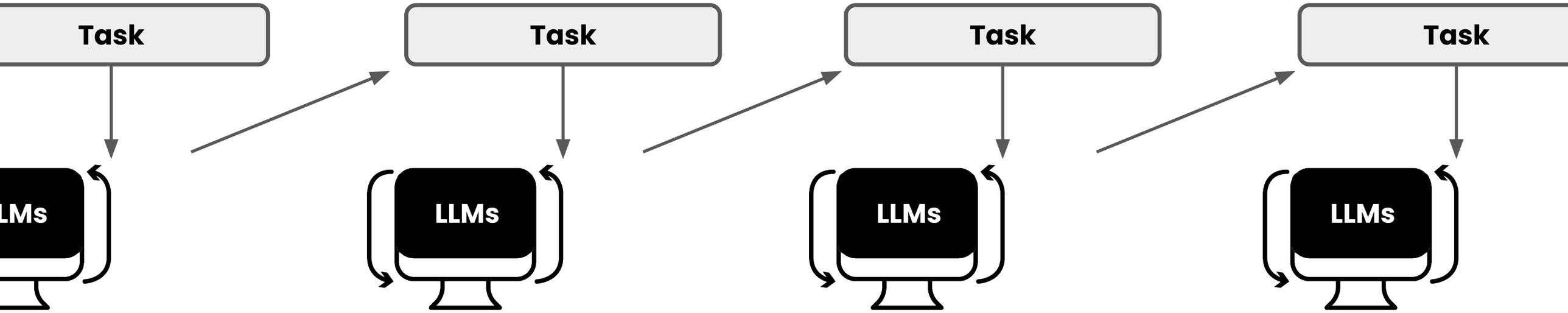


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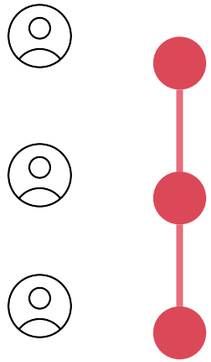
How do Multi Agents Collaborate?


Task
Task
LLMs
LLMs

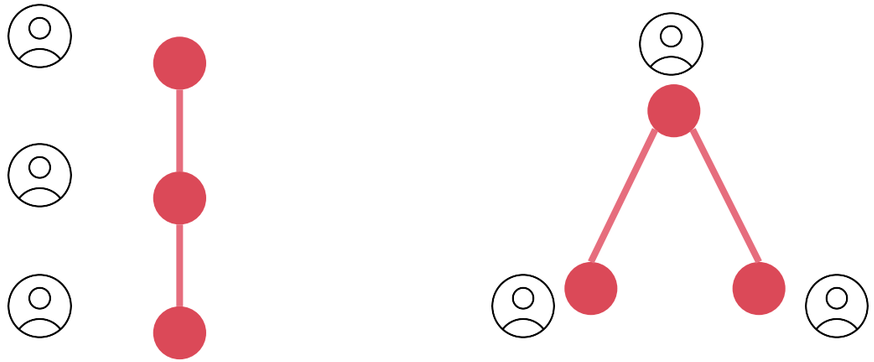

Task
Task
Task
Task
LLMs
LLMs
LLMs
LLMs


Task
Task
Task
Task
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LLMs

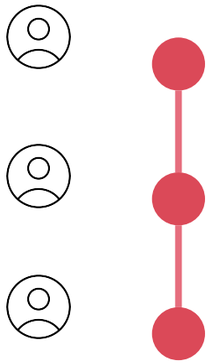
Processes

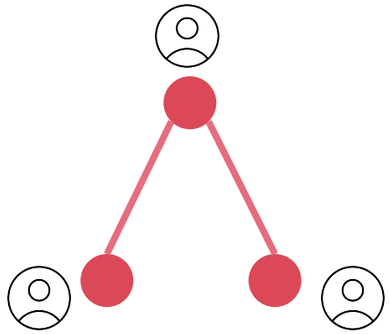


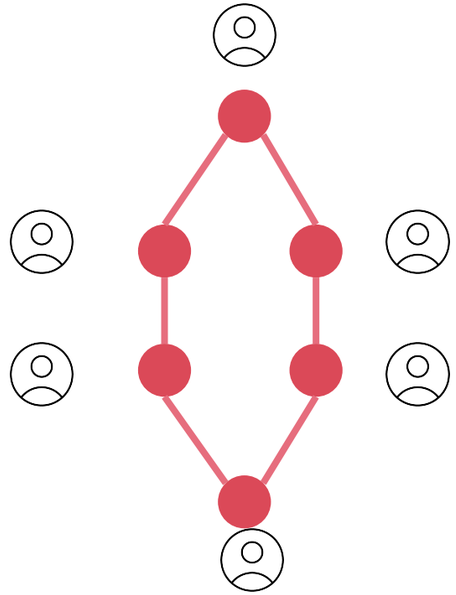
Processes



Processes



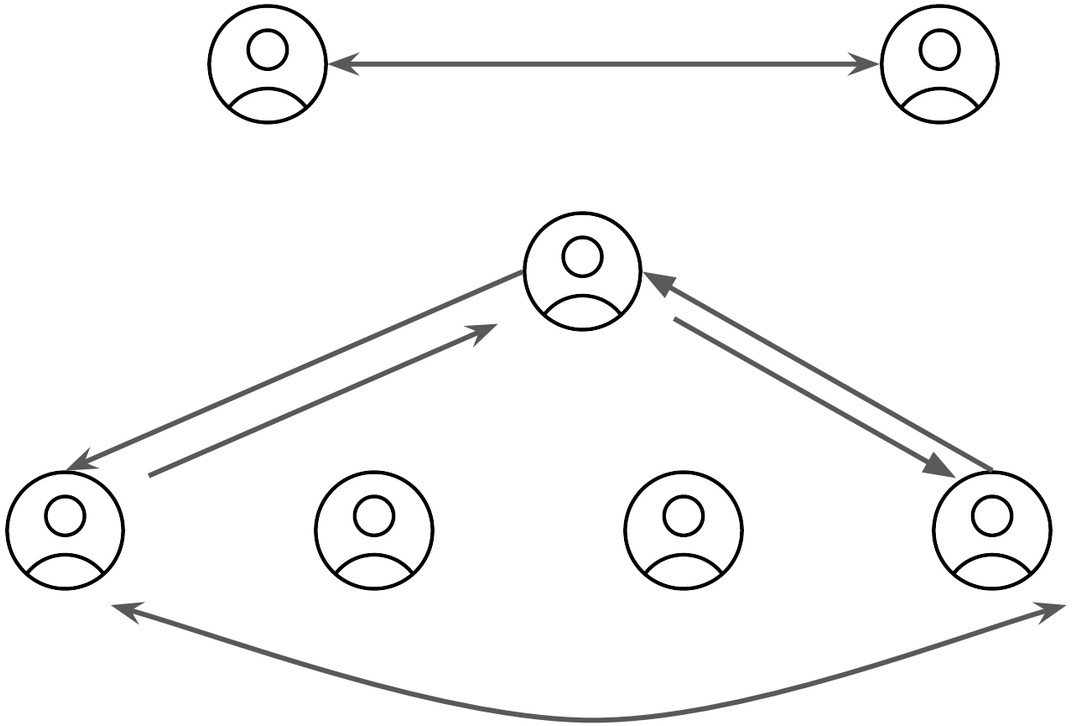


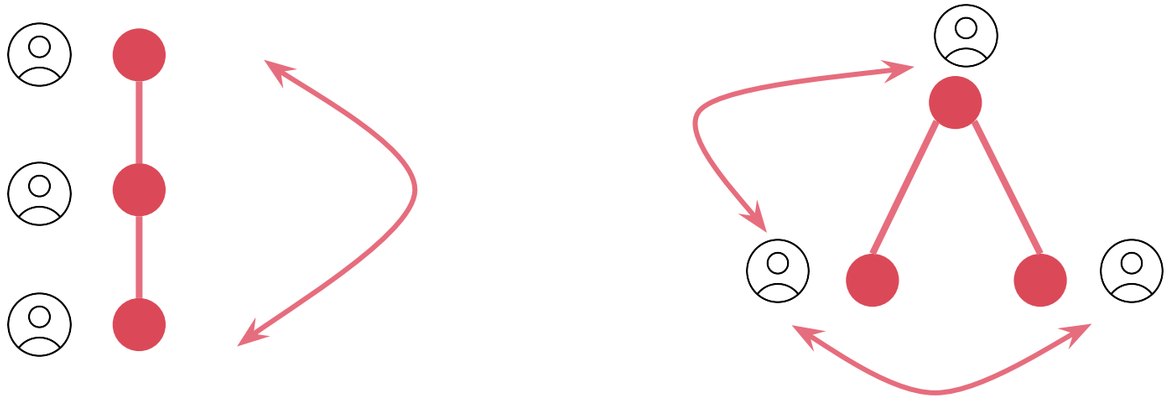


Collaboration



Collaboration





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Next steps with AI agent systems





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Conclusion





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